



JERUSALEM COLLEGE OF ENGINEERING AN AUTONOMOUS INSTITUTION

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UNIT I INTRODUCTION AND PHYSICAL LAYER

Building a network - Layered Network Protocol Architectures - Telecommunications and Cellular Networks overview - Design issues - OSI Model - TCP/IP protocol suite - Physical layer - Basics of communications - Transmission Media - Performance.

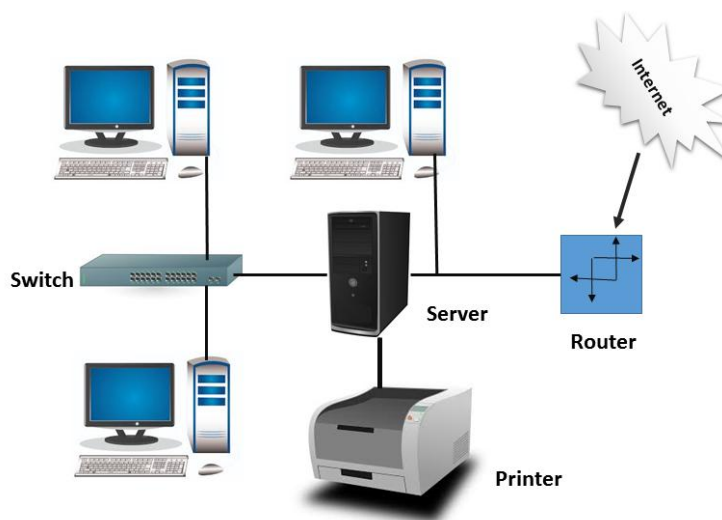
Introduction to Computer Networks

Internetwork : When two or more independent networks are connected together, the resulting network is called an internetwork or internet. Routers are responsible for connecting these independent networks and enabling communication between them.

A computer network is a collection of two or more computers and electronic devices connected together for the purpose of exchanging information and sharing resources. These devices communicate with each other through communication links such as cables, optical fibers, or wireless signals. The fundamental objective of a computer network is to enable communication and resource sharing among geographically distributed users and systems.

In the early days of computing, computers operated as standalone systems. Data had to be transferred manually using storage devices such as floppy disks or USB drives. As organizations grew and the need for information sharing increased, connecting computers became essential. This led to the development of computer networks, which allow users to communicate instantly and access shared resources from different locations.

Computer Network Diagram



Today, computer networks form the backbone of modern digital communication. The Internet itself is the world's largest computer network, connecting billions of devices across the globe. Whether sending an email, browsing a website, making an online payment, watching a video on YouTube, or participating in a virtual meeting, computer networks make these activities possible.

Importance of Computer Networks

Computer networks have become indispensable in both personal and professional environments. They **enable users to share resources** such as printers, storage devices, software applications, and Internet connections. Instead of purchasing separate resources for each user, organizations can share resources across the network, thereby reducing costs and improving efficiency.

Networks also **facilitate communication** through email, instant messaging, voice calls, and video conferencing. Employees working in different cities or countries can collaborate in real time using network-based applications. Educational institutions use networks to **provide online learning platforms**, while businesses rely on networks **for customer relationship management, inventory control, and financial transactions**.

Another important advantage of networking is reliability. Data can be backed up and stored on centralized servers, reducing the risk of information loss. Networks also support scalability, allowing organizations to add new users, devices, and services without significant modifications to the existing infrastructure.

Applications of Computer Networks

Computer networks are used in almost every sector of society. Computer networks are widely used in:

- **Education:** Online classes, virtual labs, learning management systems, and digital libraries.
- **Business:** Inventory management, customer relationship management, and enterprise communication.
- **Healthcare:** Electronic medical records, telemedicine, and hospital information systems.
- **Banking:** ATMs, online banking, mobile payments, and financial transactions.
- **Entertainment:** Netflix, YouTube, online gaming, and music streaming services.

Components of a Computer Network

Building a Network: To build a computer network, it is necessary to understand the different perspectives involved in network design and the various components that make up a network. This understanding provides the foundation for learning networking concepts and the overall structure of computer networks.

- **Application Programmer Perspective:** Application developers are concerned with the services provided by the network. Examples include reliable delivery, low delay, security, and guaranteed communication.

- **Network Operator Perspective:** Network administrators focus on manageability, fault isolation, accounting, performance monitoring, and network maintenance.

Network Designer Perspective: Network designers are interested in efficient resource utilization, scalability, fairness, and cost-effective operation.

Scalable Connectivity: A fundamental requirement of any computer network is the ability to support growth. A network that can continue to function efficiently as the number of users, devices, and applications increases is said to possess scalable connectivity. In large networks, direct communication between every pair of devices becomes impractical. Therefore, end systems are connected through intermediate devices such as switches and routers. Techniques such as switching, addressing, routing, and internetworking enable networks to expand while maintaining efficient communication and acceptable performance levels.

Cost-Effective Resource Sharing: One of the primary goals of networking is to allow multiple users to share resources in an efficient and economical manner. Resources such as communication links, storage devices, printers, software applications, and Internet connections can be shared among several users, reducing overall costs. Technologies such as multiplexing allow multiple data streams to utilize the same communication channel, thereby improving bandwidth utilization and network efficiency.

Support for Common Services: Modern applications often require similar communication services. To simplify application development, computer networks provide common services such as reliable data delivery, error detection, security, privacy, and ordered delivery of information. By providing these services at the network level, applications can focus on their specific functionality rather than implementing communication mechanisms independently.

Manageability: As networks grow in size and complexity, effective management becomes essential. Manageability refers to the ability to monitor, configure, maintain, and troubleshoot the network efficiently. Network administrators must be able to identify faults, analyze performance, manage traffic, and implement upgrades with minimal disruption. Well-designed network management systems improve reliability, scalability, and overall operational efficiency.

A computer network consists of several hardware and software components that work together to facilitate communication.

End Devices (Hosts)

End devices are the source and destination of information in a network. These include desktop computers, laptops, smartphones, tablets, servers, printers, and IoT devices. Every device connected to a network is assigned a unique address so that data can be delivered to the correct destination.



A server is a specialized computer that provides services such as file storage, web hosting, database management, or email services to other computers in the network. Client devices request services from servers and receive responses over the network.

Communication Media

Communication media refers to the physical or wireless pathways through which data travels between devices.

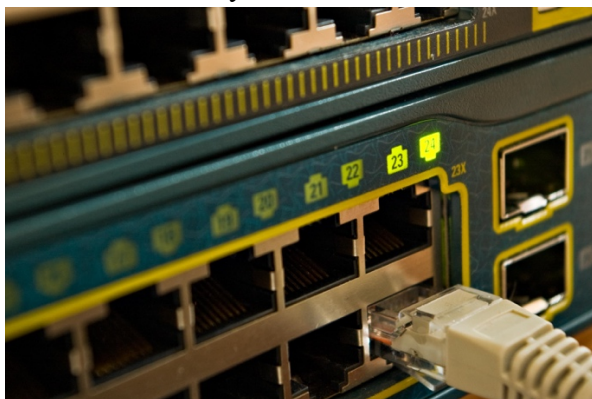
Wired media include twisted-pair cables, coaxial cables, and optical fiber cables. Twisted-pair cables are commonly used in Local Area Networks (LANs) due to their low cost and ease of installation. Optical fiber cables use light signals for communication and provide very high bandwidth and long-distance transmission capabilities.

Wireless media include radio waves, microwaves, infrared signals, and satellite communication. Wireless communication provides mobility and flexibility, allowing devices to connect without physical cables.

Networking Devices

Networking devices facilitate communication between computers and networks.

A hub is a simple device that broadcasts incoming data to all connected devices. A switch is a more intelligent device that forwards data only to the intended destination, thereby improving network efficiency.



A router connects multiple networks and determines the best path for data transmission. It is commonly used to connect local networks to the Internet. A gateway serves as a translator

between networks that use different communication protocols. Wireless access points provide Wi-Fi connectivity to wireless devices.

What is a Protocol?

For communication to occur successfully, all participating devices must follow a common set of rules. These rules are known as protocols.

A protocol is a formal set of standards and procedures that governs how data is transmitted, received, processed, and interpreted between devices in a network. Protocols define the format of messages, methods of error detection, synchronization mechanisms, addressing schemes, and procedures for establishing and terminating communication sessions.

Human communication provides a useful analogy. When two people communicate, they follow a language and certain conversational rules. Similarly, computers require protocols to understand and exchange information.

Without protocols, devices manufactured by different vendors would not be able to communicate effectively.

Common Network Protocols

- HTTP (Hypertext Transfer Protocol) is used for accessing websites and transferring web pages over the Internet. HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP and uses encryption to protect transmitted information.
- FTP (File Transfer Protocol) is used for transferring files between computers.
- SMTP (Simple Mail Transfer Protocol) is used for sending emails.
- POP3 and IMAP are used for receiving emails.
- DNS (Domain Name System) converts domain names such as www.google.com into IP addresses that computers can understand.
- TCP (Transmission Control Protocol) ensures reliable and ordered delivery of data.
- UDP (User Datagram Protocol) provides faster communication with minimal overhead but does not guarantee delivery.
- IP (Internet Protocol) is responsible for addressing and routing packets across networks.

Need for Protocol Architecture

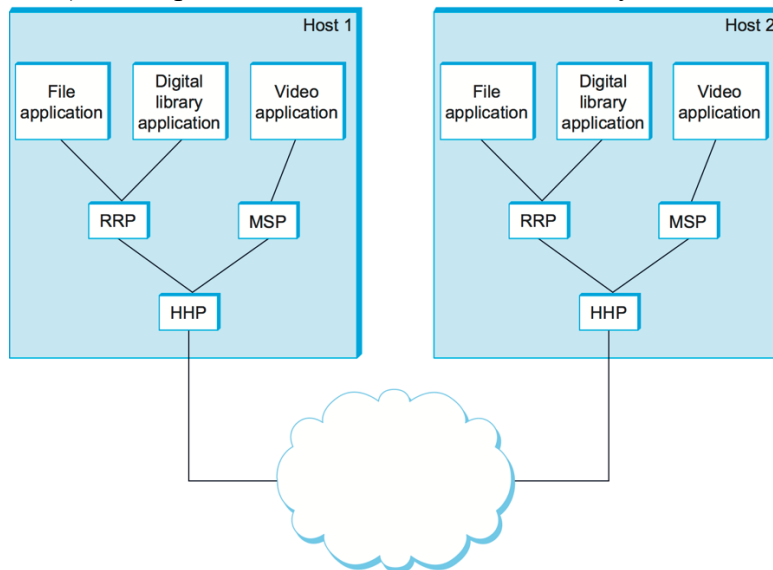
Modern network communication is highly complex. A simple activity such as opening a web page involves multiple tasks, including establishing a connection, addressing the destination, transmitting data, detecting errors, and displaying information to the user.

- Managing all these functions as a single process would be extremely difficult. Therefore, networking systems use a layered approach called protocol architecture.
- In protocol architecture, communication functions are divided into multiple layers. Each layer performs a specific task and provides services to the layer above it while receiving services from the layer below it. This modular approach simplifies network design, implementation, maintenance, and troubleshooting.
- The layered architecture ensures that changes in one layer do not significantly affect other layers. For example, improvements in physical transmission technologies can be implemented without modifying application software.

Protocol Graph

A protocol graph is a graphical representation of the protocol architecture of a network. In a protocol graph, each node represents a protocol and the edges represent dependency relationships among protocols. It illustrates how higher-level protocols depend on lower-level protocols to provide communication services

- For example, the Figure below illustrates a protocol graph consists of protocols RRP (Request/Reply Protocol) and MSP (Message Stream Protocol) implement two different types of process-to-process channels, and both depend on the HHP (Host-to-Host Protocol) which provides a host-to-host connectivity service



Layered Network Protocol Architecture

The concept of layered architecture is based on dividing communication tasks into manageable sections. Each layer is responsible for a specific set of functions and communicates with corresponding layers on remote systems.

Basic Elements of Layered Architecture

- Service: It is a set of actions that a layer provides to the higher layer.
- Protocol: It defines a set of rules that a layer uses to exchange the information with peer entity. These rules mainly concern about both the contents and order of the messages used.
- Interface: It is a way through which the message is transferred from one layer to another layer.

When a user sends data, the information moves downward through the layers of the protocol stack. Each layer adds its own control information, known as a header. At the receiving end, the data moves upward through the layers, where the headers are removed and processed.

This process is known as encapsulation and decapsulation.

The most widely recognized layered architectures are the OSI Reference Model and the TCP/IP Protocol Suite. These models provide a framework for designing and understanding network communication systems.

The layered approach offers several advantages including

- Modularity,
- Standardization,
- Interoperability,
- Easier troubleshooting, and
- Flexibility.

The layered approach decomposes the complex problem of building a network into smaller and more manageable components, thereby providing a modular design. If a new service needs to be introduced, only the corresponding layer must be modified, while the functionalities of the remaining layers can be reused. This approach also employs abstraction, which hides the underlying network complexity from the application and allows developers to focus on application-specific tasks.

As a result, it has become the foundation of modern computer networking systems.

The above topics serve as a strong foundation before introducing the **OSI Model, TCP/IP Model, Design Issues in Networks, Telecommunications Networks, and Physical Layer** in the subsequent sections of this unit.

Telecommunications and Cellular Networks Overview

Modern society depends heavily on the ability to communicate across distances. Whether a person makes a phone call, sends a text message, attends an online class, streams a video, or uses mobile banking, the underlying infrastructure that enables these services belongs to the field of telecommunications.

Telecommunications refers to the transmission of information over long distances using electronic and electromagnetic technologies. The term is derived from the Greek word *tele*, meaning "distance," and the Latin word *communicare*, meaning "to share." Telecommunications systems enable the exchange of voice, data, images, and video between users located anywhere in the world.

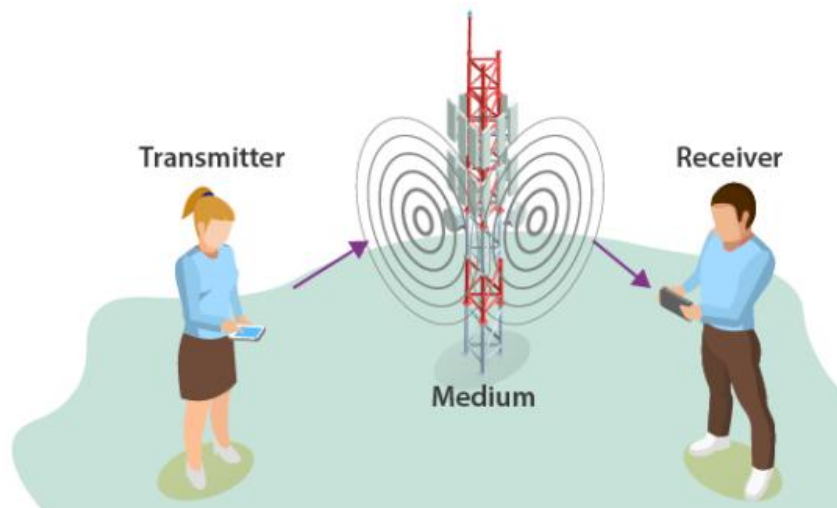
Before the development of telecommunications networks, communication over long distances relied on physical transportation of messages through letters and messengers. The invention of the telegraph, telephone, radio, and eventually computer networks revolutionized communication by allowing information to travel almost instantaneously.

Today, telecommunications forms the foundation of the Internet, mobile communication systems, satellite communication, television broadcasting, cloud computing, and countless digital services used every day.

Basic Components of a Telecommunications System

Every telecommunications system, regardless of its complexity, contains three fundamental components: a sender, a transmission medium, and a receiver.

Conceptual View of a Telecommunications System



The sender generates information that needs to be transmitted. This information may be voice, text, images, video, or data. The transmission medium carries the information from the sender to the receiver. The receiver accepts the transmitted information and converts it into a usable form.

For example, when a person makes a phone call, the smartphone acts as the sender, the cellular network acts as the transmission medium, and the recipient's phone acts as the receiver.

Evolution of Telecommunications

The history of telecommunications reflects humanity's continuous effort to improve communication speed and reach.

The telegraph was one of the earliest telecommunications systems and enabled messages to be transmitted using electrical signals. This was followed by the invention of the telephone, which allowed real-time voice communication.

The twentieth century witnessed major advancements through radio broadcasting, television, satellite communication, and digital switching systems. The emergence of the Internet transformed telecommunications by integrating voice, video, and data communication into a single digital infrastructure.

Today, telecommunications networks support billions of devices and users globally, enabling seamless communication regardless of geographical location.

Types of Telecommunications Networks

Telecommunications networks can be classified based on their coverage area and purpose.

- A Local Area Network (LAN) connects devices within a limited geographical area such as a home, office, or campus.
- A Metropolitan Area Network (MAN) connects multiple LANs within a city or metropolitan region.
- A Wide Area Network (WAN) spans large geographical regions, connecting cities, countries, and continents. The Internet is the largest example of a WAN.
- A SAN (Storage Area Network) is a specialized high-speed network that connects servers to storage devices. It is generally confined to a data center or a single room and provides centralized storage management. SANs are commonly used in cloud

environments, enterprise data centers, banking systems, and high-performance computing environments.

Modern telecommunications systems often integrate these network types to provide global connectivity.

Line Configuration

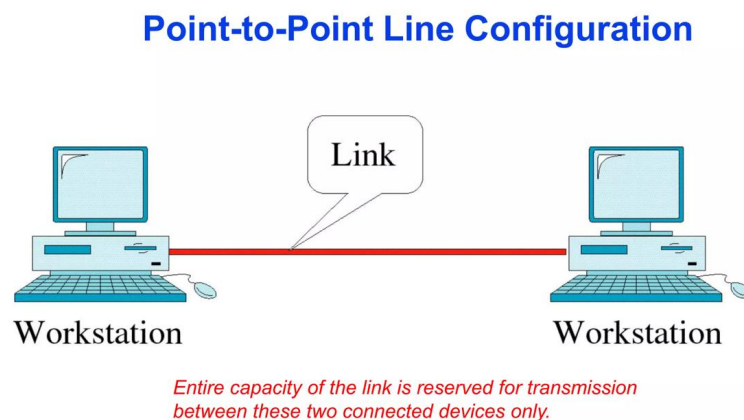
Before devices can communicate, a communication link must be established between them. The manner in which two or more devices are connected through a communication link is known as **line configuration**. A Line configuration determines how communication resources are shared and how devices access the transmission medium.

In computer networks, two fundamental line configurations are used:

1. Point-to-Point Configuration
2. Multipoint Configuration

Point-to-Point Configuration: A point-to-point connection provides a dedicated communication link between exactly two devices. The entire capacity of the communication channel is reserved for those two devices.

Point-to-Point Communication



When one device transmits data, the other device receives it directly through the dedicated link. A simple example is a direct connection between two computers using a cable.

Telephone communication is another example of point-to-point communication because the communication path is dedicated to the participants during the call.

Characteristics

Since the communication channel is dedicated, point-to-point connections generally provide better security, higher reliability, and predictable performance.

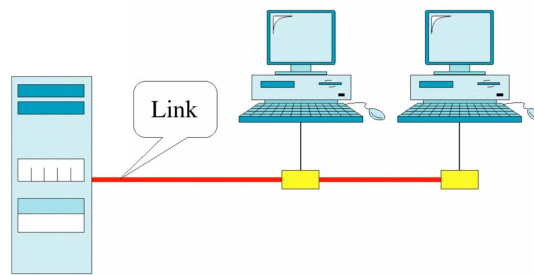
However, establishing separate links between every pair of devices becomes expensive in large networks.

Multipoint Configuration

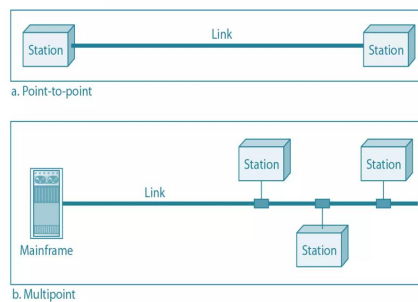
In a multipoint configuration, multiple devices share a single communication link. The capacity of the link is shared among all participating devices.

Multipoint Communication

Multipoint Line Configuration



Comparison of point to Point & Multi-Point Line Configuration



For example, in a wireless network, many devices communicate through a common access point and share the available bandwidth.

The efficiency of a multipoint system depends on how effectively the shared medium is managed.

Characteristics

Multipoint configurations are cost-effective and easier to expand. However, as the number of users increases, congestion and performance degradation may occur.

Network Topology

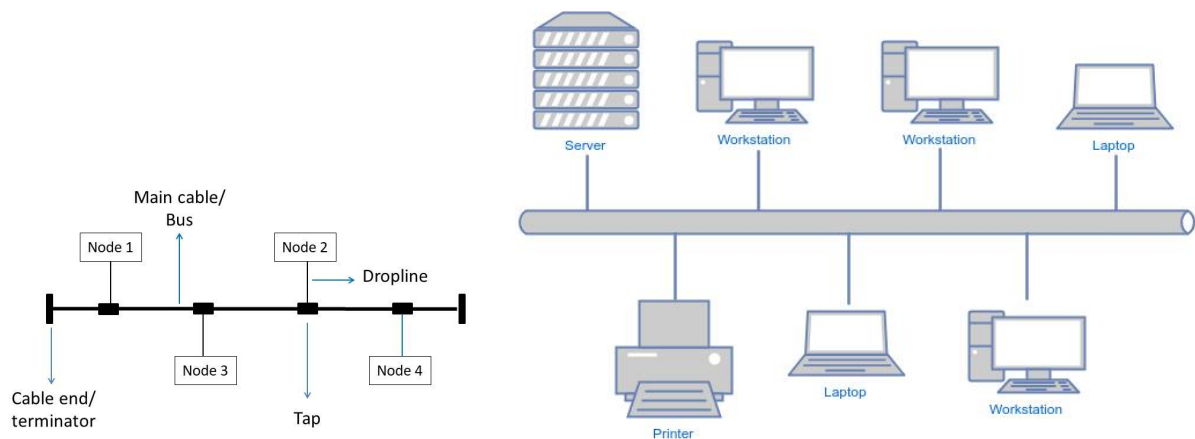
After understanding how devices share communication links, the next question is how those devices are physically arranged within a network. The physical or logical arrangement of devices and communication links in a network is called **network topology**.

Topology determines how devices communicate, how data flows through the network, and how failures affect network operation.

Choosing an appropriate topology is one of the most important decisions in network design.

1. Bus Topology

In a bus topology, all devices are connected to a single communication cable known as the backbone. Every device shares the same transmission medium.



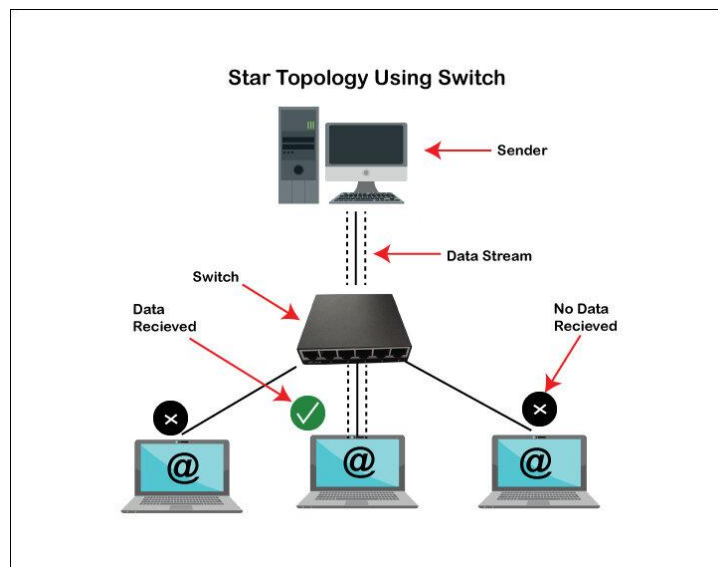
When a device transmits data, the signal travels along the backbone and reaches all connected devices. Only the intended recipient accepts the message.

Advantages: Bus topology requires less cabling and is relatively inexpensive to implement.

Limitations: If the backbone cable fails, the entire network becomes inoperative. Performance also decreases as the number of devices increases.

2. Star Topology

In a star topology, every device is connected to a central networking device such as a hub or switch. All communication passes through the central device.



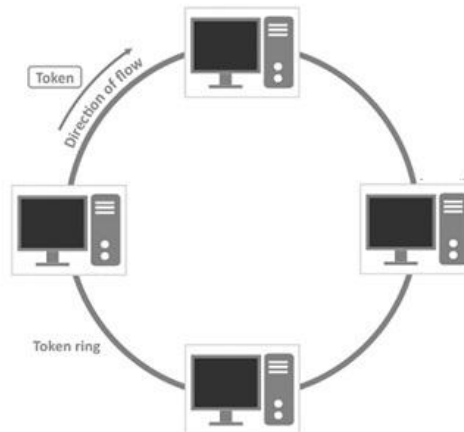
Modern Ethernet networks predominantly use star topology.

Advantages: The failure of one communication link affects only the corresponding device and does not disrupt the entire network. Troubleshooting and maintenance are easier.

Limitations: If the central switch or hub fails, the entire network becomes unavailable.

3. Ring Topology

In a ring topology, devices are connected in a circular arrangement. Each device is connected to exactly two neighboring devices.



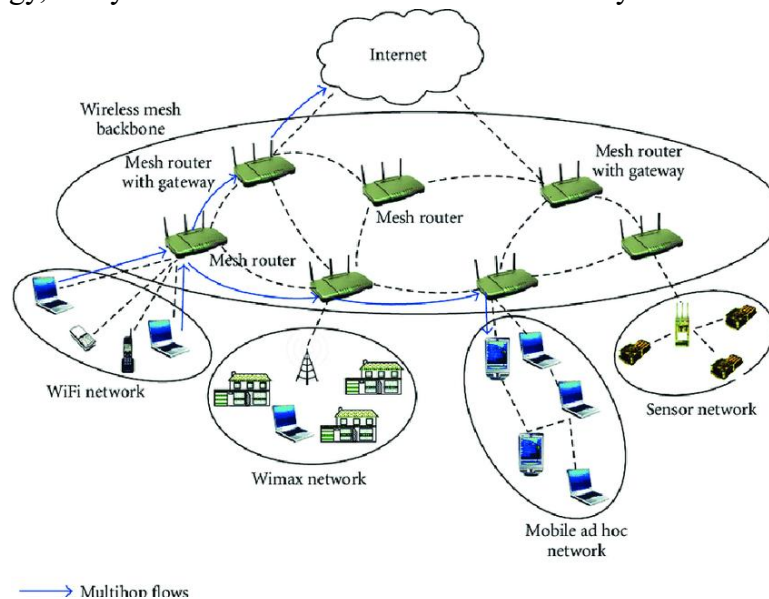
Data travels from one device to another around the ring until it reaches its destination.

Advantages: Ring topology provides orderly communication and reduces data collisions.

Limitations: A failure in one device or communication link can affect the entire network unless redundancy mechanisms are implemented.

4. Mesh Topology

In a mesh topology, every device has a direct connection to every other device.



Multiple communication paths exist between devices.

Advantages: Mesh topology provides excellent reliability, fault tolerance, and redundancy. Even if several links fail, communication can continue through alternative paths.

Limitations: The number of communication links increases dramatically as the number of devices grows, making implementation expensive.

Mesh Formula (Important Numerical)

For n devices:

$$\text{Number of Links} = \frac{n(n-1)}{2}$$

Example

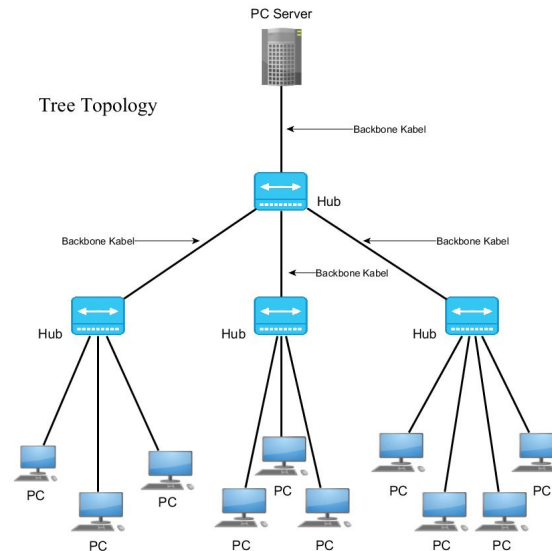
For 5 devices:

$$\frac{5(5-1)}{2} = 10 \text{ links are required.}$$

This is one of the few numericals directly associated with network topology.

5. Tree Topology

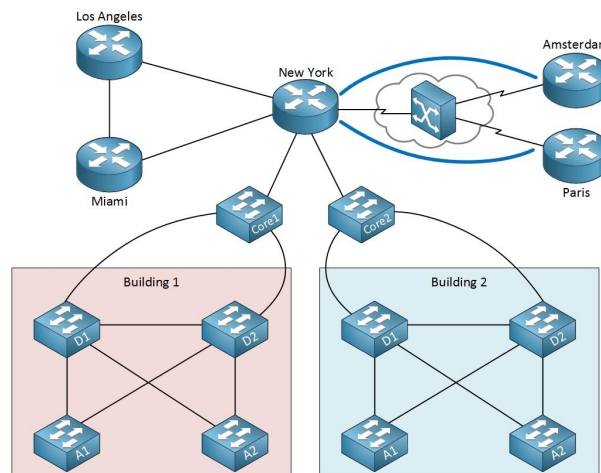
A tree topology combines characteristics of bus and star topologies. It organizes devices in a hierarchical structure resembling a tree.



Large organizations frequently use tree topology because it supports hierarchical management and easy network expansion.

6. Hybrid Topology

Modern networks rarely use a single topology. Instead, they combine multiple topologies to achieve better performance and flexibility. Such networks are known as hybrid networks.



Most enterprise networks today are hybrid in nature, combining star, mesh, and tree structures.

Cellular Networks

Why Cellular Networks Were Introduced

Traditional telephone networks relied on wired connections between users. Although effective, wired systems limited mobility because users had to remain connected to physical communication lines.

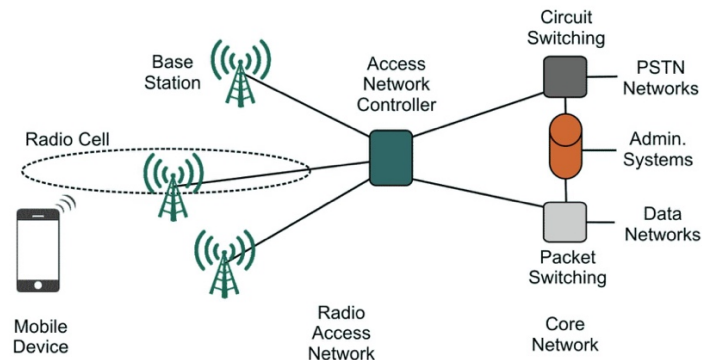
As mobile communication became necessary, engineers developed cellular networks. Cellular networks allow users to communicate while moving from one location to another without losing connectivity.

The term *cellular* originates from the division of a geographical region into multiple smaller areas called cells. Each cell contains a base station responsible for providing wireless communication services to devices within its coverage area.

Concept of a Cell

A cellular network divides a large geographical area into smaller cells.

Cellular Network Structure



Each cell contains a Base Transceiver Station (BTS), commonly known as a cell tower. Mobile devices communicate with the nearest base station using radio waves.

As users move from one cell to another, the communication session is transferred automatically from one base station to another. This process is called **handoff** or **handover**.

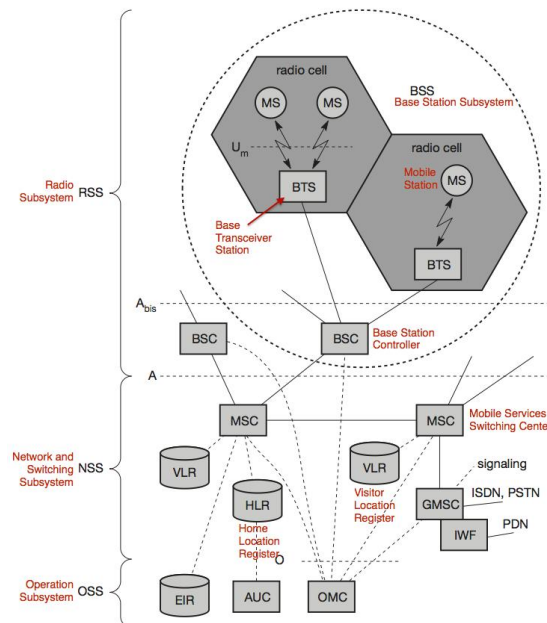
The cellular architecture allows frequency reuse, which significantly increases network capacity. Instead of assigning unique frequencies to every user, frequencies can be reused in non-adjacent cells without causing interference.

Components of a Cellular Network

A cellular network consists of several interconnected components.

- **Mobile Station (MS):** The mobile station refers to the user's device, such as a smartphone or tablet. It is responsible for transmitting and receiving signals.
- **Base Transceiver Station (BTS):** The BTS provides radio communication between mobile devices and the cellular network. It manages signal transmission within a specific cell.
- **Base Station Controller (BSC):** The BSC manages multiple base stations and coordinates handovers between cells.
- **Mobile Switching Center (MSC):** The MSC acts as the central controller of the cellular network. It handles call routing, authentication, billing, and mobility management.

Cellular Network Architecture



Together, these components ensure uninterrupted communication as users move through different geographical areas.

Generations of Cellular Networks

The development of cellular technology is commonly described in terms of generations.

- **First Generation (1G):** The first generation of cellular systems used analog technology. These systems primarily supported voice communication and suffered from poor security and limited capacity.
- **Second Generation (2G):** The second generation introduced digital communication techniques. It improved voice quality, increased capacity, and introduced Short Message Service (SMS).
- **Third Generation (3G):** Third-generation systems provided higher data transmission rates, enabling mobile Internet access, video calls, and multimedia applications.
- **Fourth Generation (4G):** Fourth-generation networks significantly increased bandwidth and reduced latency. Technologies such as LTE enabled high-definition video streaming and fast Internet connectivity.
- **Fifth Generation (5G):** Fifth-generation networks provide ultra-high speeds, extremely low latency, and support for billions of Internet of Things (IoT) devices.

Evolution of Cellular Networks

The transition from 1G to 5G illustrates how telecommunications systems have evolved from simple voice communication networks into sophisticated digital ecosystems.

Design Issues in Computer Networks

When designing a network, engineers must ensure that communication occurs efficiently, accurately, securely, and reliably. Regardless of whether the network is a small office LAN, a

cellular network, or the global Internet, several fundamental design challenges must be addressed.

These challenges are known as **design issues in computer networks**.

A well-designed network must answer questions such as:

- How will devices identify each other?
- How will errors be detected and corrected?
- How will congestion be controlled?
- How will security be maintained?
- How will data reach the correct destination?

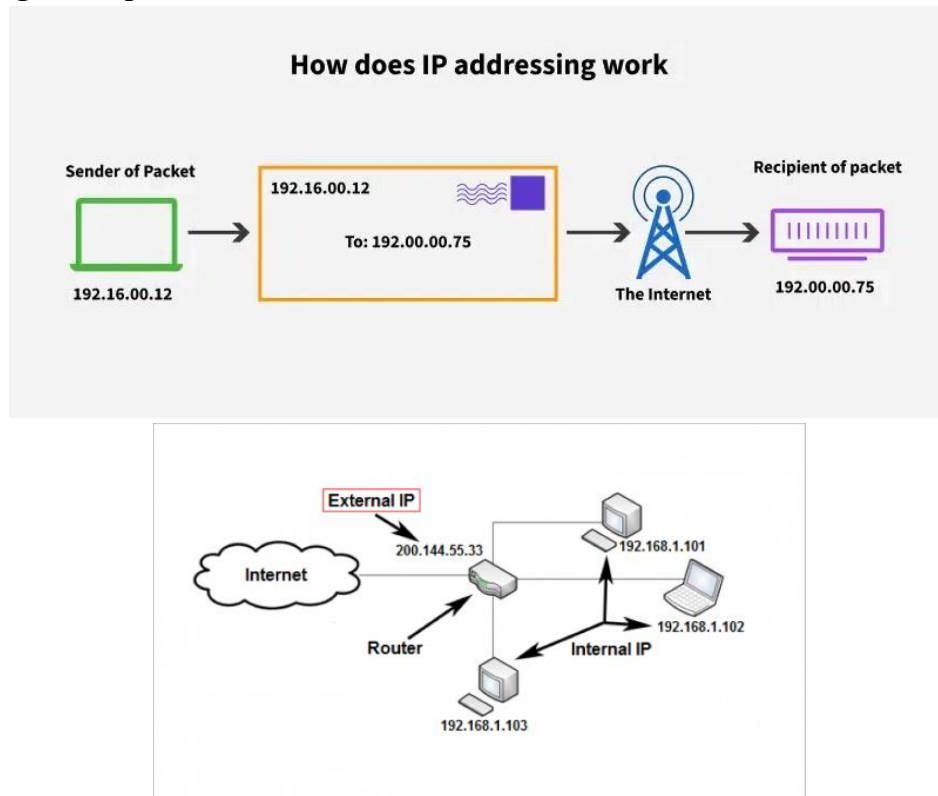
Addressing these questions forms the basis of network design.

Addressing

Every device connected to a network must possess a unique identity. Without addressing, data would have no way of reaching its intended destination.

Consider a postal system. A letter can only be delivered if it contains the correct address. Similarly, network devices require addresses to exchange information.

Addressing Concept



Internet Protocol (IP) addresses serve as logical addresses that uniquely identify devices on a network.

Routing

- Once devices have addresses, the network must determine the path through which data should travel.
- Routing refers to the process of selecting the most efficient path between source and destination.

- Routers maintain routing tables and use routing algorithms to determine optimal paths for packet transmission.
- Without routing, communication across large networks such as the Internet would be impossible.

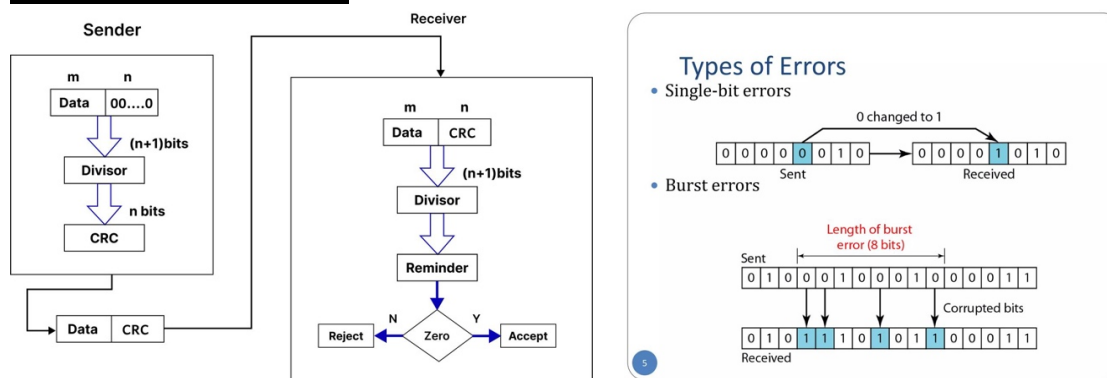
Error Control

Data transmission is vulnerable to noise, interference, attenuation, and hardware failures. These factors can corrupt transmitted information.

Error control mechanisms ensure that errors are detected and corrected before the data reaches the application.

Common techniques include parity checks, checksums, cyclic redundancy checks (CRC), acknowledgments, and retransmissions.

Error Detection Concept



Error control is essential for maintaining communication reliability.

Flow Control

A sender may transmit data faster than a receiver can process it.

Flow control mechanisms regulate the transmission rate to prevent receiver overload and data loss. Protocols such as TCP implement flow control using sliding window mechanisms.

Flow control ensures smooth communication between devices with different processing capabilities.

Congestion Control

Network congestion occurs when the volume of traffic exceeds available network capacity.

Congestion can lead to packet loss, increased delays, and reduced throughput.

Congestion control techniques monitor network traffic and adjust transmission rates to maintain efficient communication.

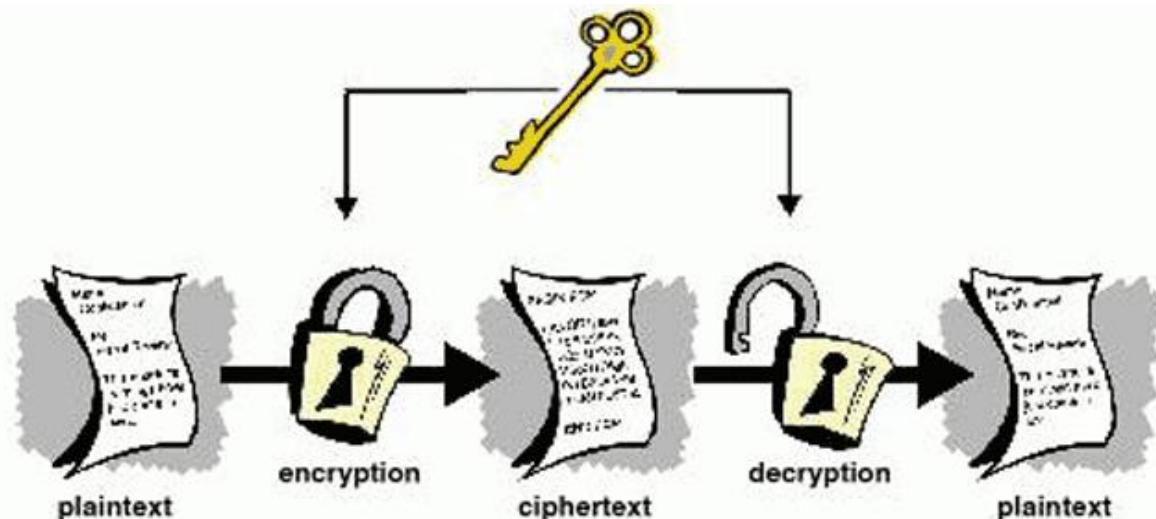
This issue becomes especially important in high-traffic environments such as data centers and cellular networks.

Security: Security is one of the most critical design issues in modern networks.

Networks must protect information from unauthorized access, modification, and destruction.

Security mechanisms include authentication, encryption, digital signatures, firewalls, and intrusion detection systems.

Network Security Concept



Without adequate security, sensitive information such as passwords, financial records, and personal data would be vulnerable to cyberattacks.

Telecommunications provides the technological foundation for long-distance communication, while cellular networks enable mobile communication through the concept of cells, base stations, and frequency reuse. Modern cellular systems have evolved from 1G analog networks to advanced 5G digital infrastructures capable of supporting massive connectivity and high-speed data services.

The successful operation of these networks depends on solving fundamental design issues such as addressing, routing, error control, flow control, congestion control, and security. These design issues are not limited to cellular networks; they are essential considerations in every computer network, from small local networks to the global Internet. Understanding these concepts provides the foundation for studying protocol architectures such as the OSI Model and TCP/IP Protocol Suite in subsequent topics.

OSI Model

Whenever a user sends a WhatsApp message, browses a website, uploads a file to Google Drive, or streams a video on YouTube, an enormous amount of communication occurs behind the scenes. The message travels through multiple devices, communication media, protocols, and networks before reaching its destination.

For communication to occur successfully, different devices manufactured by different companies must follow common standards. A smartphone manufactured by Samsung should be able to communicate with a server manufactured by Dell. Similarly, a Windows computer should be able to communicate with a Linux server. This interoperability is achieved through standardized communication models and protocols.

Two of the most important frameworks used in networking are the **OSI Reference Model** and the **TCP/IP Protocol Suite**. These models divide the communication process into layers, making network design, implementation, and troubleshooting much easier.

Before studying these models, it is important to understand the fundamentals of communication and the challenges involved in transmitting information across networks.

Basics of Communication

What is Communication?

Communication is the process of transferring information from one entity to another. In computer networks, communication refers to the exchange of data between devices through a transmission medium.

The information being communicated may be text, audio, video, images, sensor readings, or any other form of digital data.

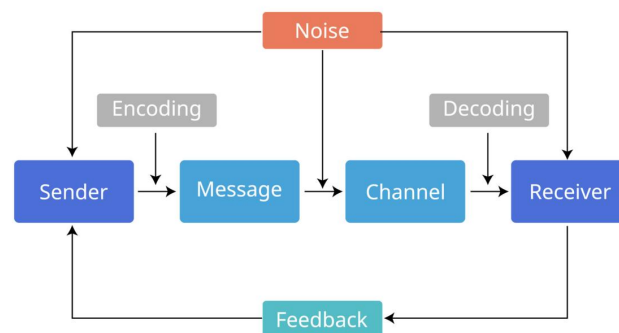
For communication to be successful, the sender and receiver must understand the same rules and conventions.

Elements of a Data Communication System

Every communication system consists of five fundamental components.

Basic Communication System

The Communication Process



Sender: The sender is the device that generates and transmits information.

Examples include:

- Computer
- Smartphone
- Server
- Sensor

Receiver: The receiver is the destination device that accepts and processes the transmitted information.

Examples include:

- Laptop
- Mobile phone
- Printer
- Database server

Message: A message is the information being transmitted.

Examples include:

- Text
- Images
- Audio
- Video
- Documents

Transmission Medium: The transmission medium is the path through which data travels.

Examples include:

- Twisted pair cable
- Optical fiber

- Radio waves
- Satellite links

Protocol: A protocol is a set of rules governing communication between devices.

Protocols define:

- Message format
- Transmission speed
- Error handling
- Addressing methods
- Session management

Without protocols, communication between devices would be impossible.

Characteristics of Effective Data Communication

A communication system is considered effective when it satisfies four important requirements.

- **Delivery:** Data must reach the correct destination.
- **Accuracy:** Data must arrive without errors.
- **Timeliness:** Data should arrive within an acceptable time period.
- **Jitter:** Variation in packet arrival times should be minimized, especially in audio and video applications.

Need for Layered Architecture

As communication networks became increasingly complex, managing all communication functions within a single software system became impractical.

Consider opening a website:

The system must:

- Establish a connection
- Identify the destination
- Route packets
- Detect errors
- Control transmission speed
- Format data
- Display information to the user

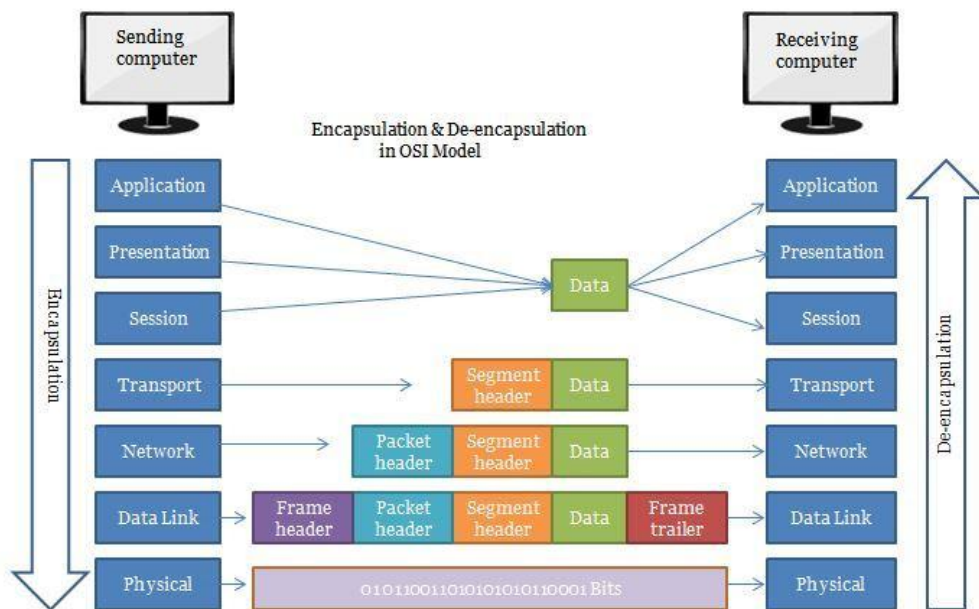
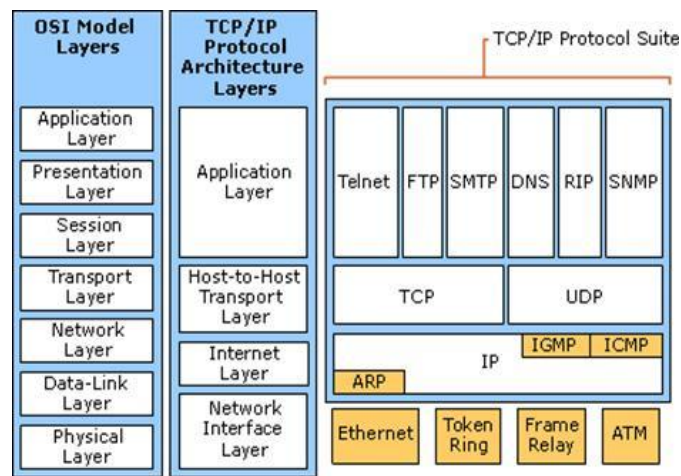
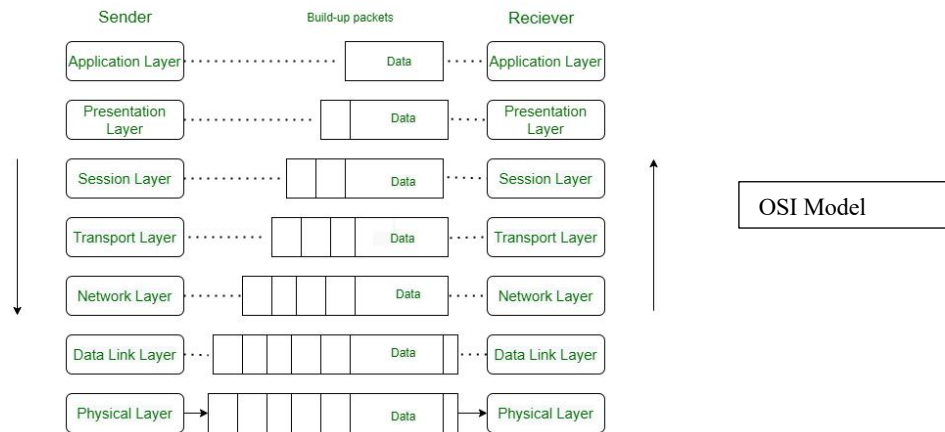
If all these functions were handled together, network design would become extremely difficult.

To overcome this challenge, communication tasks are divided into layers.

Layered Communication Concept

The layered approach organizes communication functions into separate levels, where each level performs a specific task.

Layered Communication



Each layer provides services to the layer above it and receives services from the layer below it. This approach offers:

- Modularity
- Standardization
- Easier maintenance

- Better interoperability
- Simplified troubleshooting

The most famous layered architectures are the OSI Model and the TCP/IP Protocol Suite.

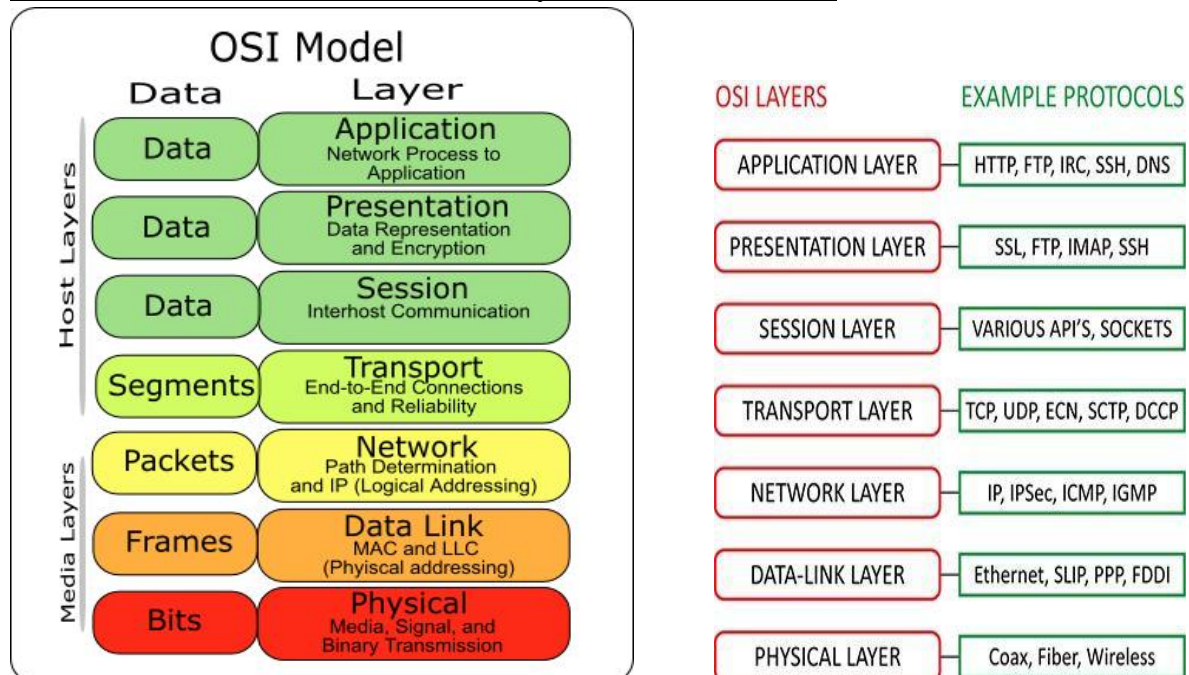
OSI Reference Model

The Open Systems Interconnection (OSI) model was developed by the International Organization for Standardization (ISO) to provide a standardized framework for network communication.

The OSI model divides communication into seven layers, each responsible for a specific set of functions.

The objective of the OSI model is not to define protocols but to provide a conceptual framework for understanding communication processes.

Structure of the OSI Model - Seven Layers of the OSI Model



The seven layers are:

1. Application Layer
2. Presentation Layer
3. Session Layer
4. Transport Layer
5. Network Layer
6. Data Link Layer
7. Physical Layer

Communication occurs from the Application Layer down to the Physical Layer at the sender's side and from the Physical Layer up to the Application Layer at the receiver's side.

Application Layer (Layer 7) - The Application Layer is the layer closest to the user.

It provides network services directly to applications and enables users to interact with network resources.

Examples include:

- Web browsing
- Email services

- File transfer
- Remote login

Common protocols include HTTP, HTTPS, FTP, SMTP, DNS, and Telnet.

Whenever a user opens a web browser or sends an email, communication begins at this layer.

Presentation Layer (Layer 6) - Different computer systems may represent data in different formats. The Presentation Layer ensures that data is translated into a common format before transmission.

Its primary responsibilities include:

- Data translation
- Data compression
- Data encryption
- Data decryption

For example, HTTPS communication uses encryption techniques that operate at this layer.

Session Layer (Layer 5) - The Session Layer establishes, manages, and terminates communication sessions between applications.

A session is a logical conversation between two devices.

Functions include:

- Session establishment
- Session maintenance
- Synchronization
- Session termination

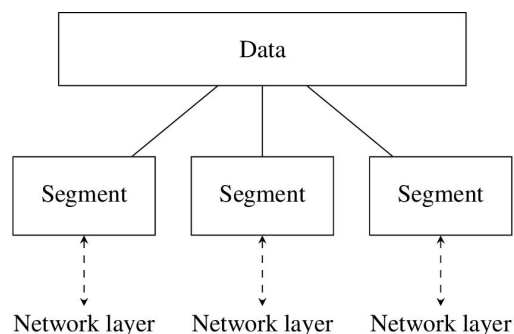
Video conferencing applications frequently use session management techniques.

Transport Layer (Layer 4) - The Transport Layer ensures reliable end-to-end communication between sender and receiver.

Its responsibilities include:

- Segmentation
- Flow control
- Error control
- Reliable delivery

Transport Layer Communication



The differences between TCP and UDP

Transmission Control Protocol	User Datagram Protocol
A connection-oriented protocol	A connectionless protocol
The most widely used protocol on the internet	Used for voice over IP, streaming video, gaming and live broadcasts
Guarantees that no packets are missing and all the data that's sent makes it to the intended recipient	Faster and needs fewer resources
Sends packets in order so they can be stitched back together easily	Packets don't necessarily arrive in order
Slower and requires more resources	Allows missing packets, but the sender is unable to know whether a packet has been received
Has a bigger header than UDP, best suited for apps that need high reliability and transmission time is relatively less critical	Better suited for applications that need fast, efficient transmission, such as games

Two important protocols operate at this layer:

TCP (Transmission Control Protocol) - TCP provides reliable communication through acknowledgments, sequencing, retransmissions, and error recovery. Applications such as email, file transfer, and web browsing typically use TCP.

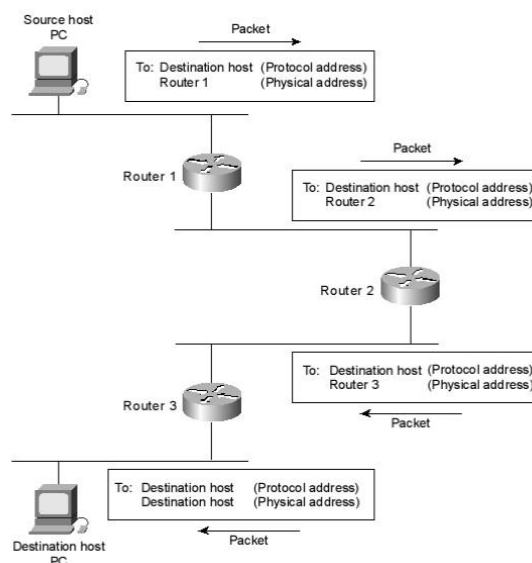
UDP (User Datagram Protocol) - UDP provides faster communication by eliminating reliability overhead. Applications such as video streaming, gaming, and live broadcasting often use UDP.

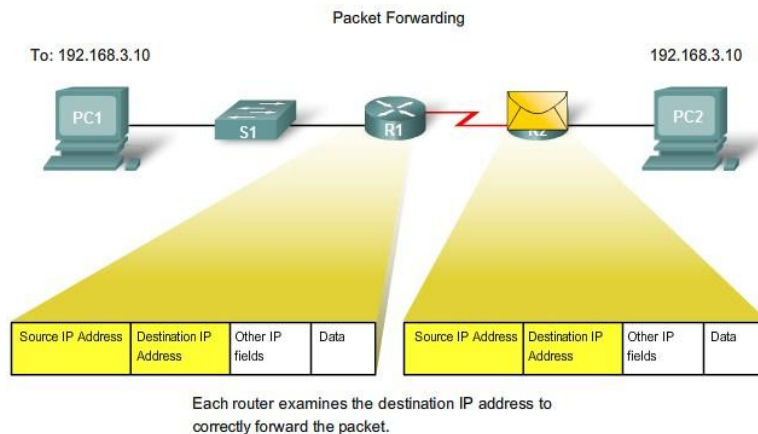
Network Layer (Layer 3) - The Network Layer is responsible for routing packets between networks.

Its primary functions include:

- Logical addressing
- Routing
- Path determination
- Packet forwarding

The Internet Protocol (IP) operates at this layer. Routers primarily function at the Network Layer.





Data Link Layer (Layer 2) - The Data Link Layer provides node-to-node communication over a physical link.

Its functions include:

- Framing
- Error detection
- Flow control
- MAC addressing

Switches primarily operate at this layer. The Data Link Layer ensures reliable transmission between directly connected devices.

Physical Layer (Layer 1) - The Physical Layer is responsible for transmitting raw bits through the communication medium. It deals with electrical, optical, mechanical, and procedural characteristics of communication.

Examples include:

- Ethernet cables
- Fiber optic cables
- Radio waves
- Connectors
- Voltage levels

The Physical Layer does not interpret data; it simply transmits bits. Because of its importance, it deserves separate study.

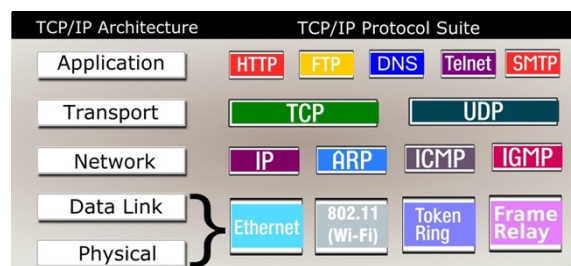
TCP/IP Protocol Suite

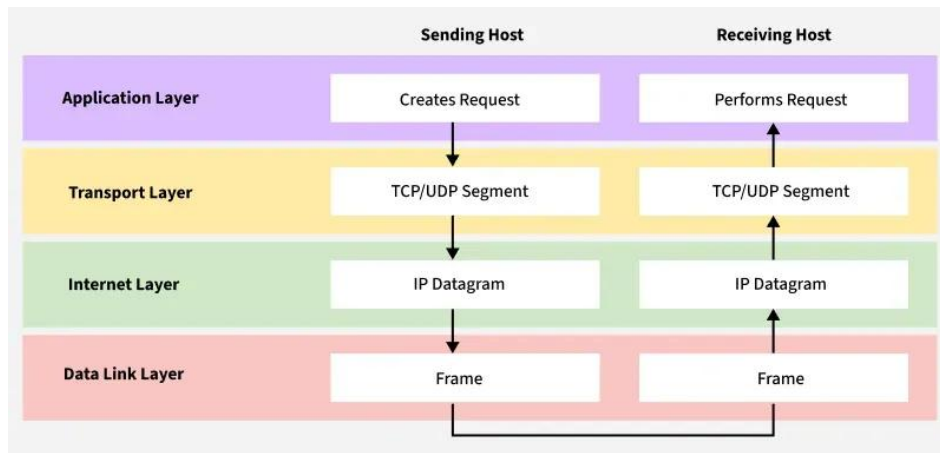
Although the OSI model is widely used for understanding networking concepts, the Internet actually operates using the TCP/IP Protocol Suite.

The TCP/IP model was developed by the United States Department of Defense and became the foundation of the modern Internet.

Unlike the OSI model, TCP/IP is both a conceptual model and a practical protocol architecture.

Structure of TCP/IP Protocol Suite - TCP/IP Architecture





The TCP/IP model contains four layers:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Access Layer

Application Layer - This layer combines the functions of the OSI Application, Presentation, and Session layers.

Protocols include:

- HTTP
- HTTPS
- SMTP
- FTP
- DNS
- Telnet

This layer provides services directly to users and applications.

Transport Layer - The Transport Layer provides end-to-end communication services. The two major protocols are:

1. TCP Provides:

- Reliable communication
- Error recovery
- Flow control
- Sequencing

2. UDP Provides:

- Fast communication
- Low overhead
- Minimal delay

Internet Layer - The Internet Layer is responsible for logical addressing and routing.

Important protocols include:

- IP
- ICMP
- ARP

This layer ensures that packets reach the correct destination across interconnected networks.

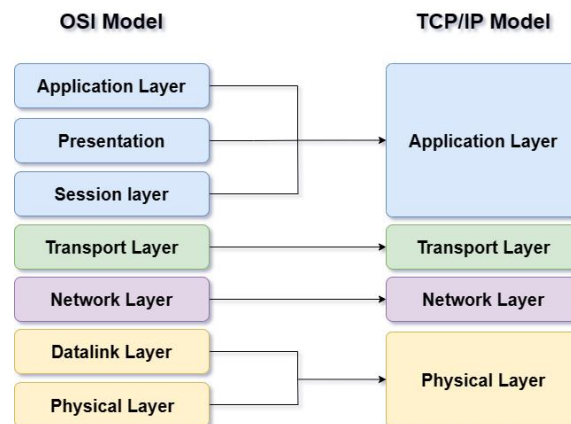
Network Access Layer - This layer combines the functions of the OSI Data Link and Physical layers. Its responsibilities include:

- Frame transmission
- Physical addressing
- Media access control
- Physical signal transmission

Examples include Ethernet and Wi-Fi technologies.

OSI vs TCP/IP

Comparison of Models



S.No	OSI MODEL	TCP/IP MODEL
1	Defined before advent of internet	Defined after the advent of Internet.
2	Service interface and protocols are clearly distinguished before	Service interface and protocols were not clearly distinguished before
3	Internetworking not supported	TCP/IP supports Internet working
4	Strict layering	Loosely layered
5	Protocol independent standard	Protocol Dependant standard
6	Less Credible	More Credible
7	All packets are reliably delivered	TCP reliably delivers packets, IP does not reliably deliver packets

The OSI model contains seven layers and serves primarily as a reference framework. The TCP/IP model contains four layers and forms the practical foundation of the Internet.

Most modern networking technologies follow the TCP/IP architecture while still using OSI terminology for explanation and troubleshooting.

Physical Layer

The Physical Layer is the lowest layer of the OSI model and serves as the foundation of all network communication. Its primary responsibility is to transmit raw binary data (0s and 1s) through the communication medium.

The Physical Layer does not understand the meaning of data. It simply converts bits into signals and transmits them across the network.

Functions of the Physical Layer

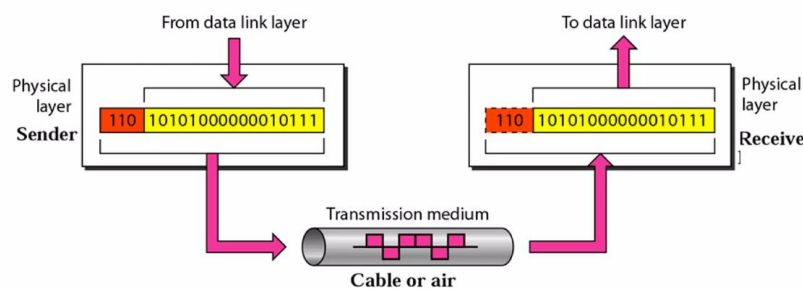
Physical Layer Functions

Functions of Physical Layer

1. Bit-by-Bit Transmission
2. Encoding and Decoding
3. Signal Transmission
4. Modulation and Demodulation
5. Transmission Modes
6. Data Control

Physical Media

Physical Media



The physical layer is responsible for movements of individual bits from one node to the next.

The Physical Layer is responsible for:

- Representation of bits
- Data rate specification
- Bit synchronization
- Line configuration
- Physical topology
- Transmission mode

Transmission Modes - Communication may occur in three different modes.

- **Simplex Communication:** Data flows in only one direction.

Example:

Keyboard → Computer

- **Half Duplex Communication:** Data flows in both directions but not simultaneously.

Example:

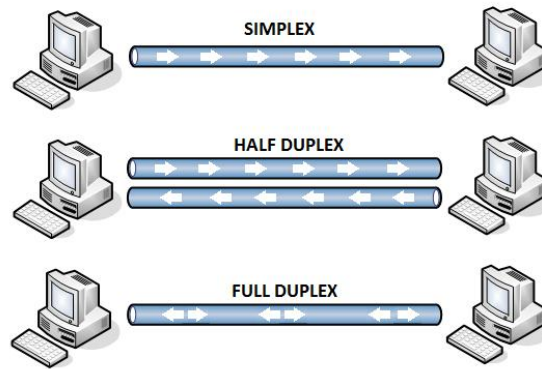
Walkie-talkie

- **Full Duplex Communication:** Data flows simultaneously in both directions.

Example:

Telephone conversation

Transmission Modes



Encapsulation and Decapsulation

One of the most important concepts in networking is encapsulation.

When data moves down the protocol stack, each layer adds its own header containing control information. At the receiving side, each layer removes its corresponding header.

Encapsulation Process

PDU (Protocol Data Unit) Names of the Layers of the OSI Model

OSI Model Layers	PDU Names
Application	PDU
Presentation	PDU
Session	PDU
Transport	Segment
Network	Packet
Data Link	Frame
Physical	Bits

This process allows layers to communicate independently while maintaining overall network functionality.

Communication in computer networks begins with the basic exchange of information between a sender and a receiver through a transmission medium using agreed-upon protocols. To manage the complexity of communication, networking systems adopt layered architectures.

The **OSI Model** provides a seven-layer conceptual framework for understanding communication functions, while the **TCP/IP Protocol Suite** provides the practical architecture used by the Internet. At the foundation of both models lies the **Physical Layer**, which converts digital information into signals and transmits them through communication media. Together, these concepts form the core framework upon which all modern computer networks operate.

Transmission Media and Network Performance

In the previous sections, we studied how communication occurs between devices, the role of protocols, the layered architecture of networks, the OSI model, the TCP/IP protocol suite, and the Physical Layer. However, regardless of how sophisticated a protocol may be,

communication cannot occur unless there exists a path through which data can travel from the sender to the receiver.

This path is known as the **transmission medium**.

Whenever an email is sent, a video is streamed, a mobile phone call is made, or a web page is loaded, data must physically travel through some medium. The quality, speed, reliability, and efficiency of communication depend heavily on the characteristics of this medium.

Transmission media form the foundation of the Physical Layer because they provide the actual channel through which signals travel. Understanding transmission media is therefore essential for understanding how networks function in the real world.

What is Transmission Media?

Transmission media refers to the physical or wireless pathway that carries information from the sender to the receiver.

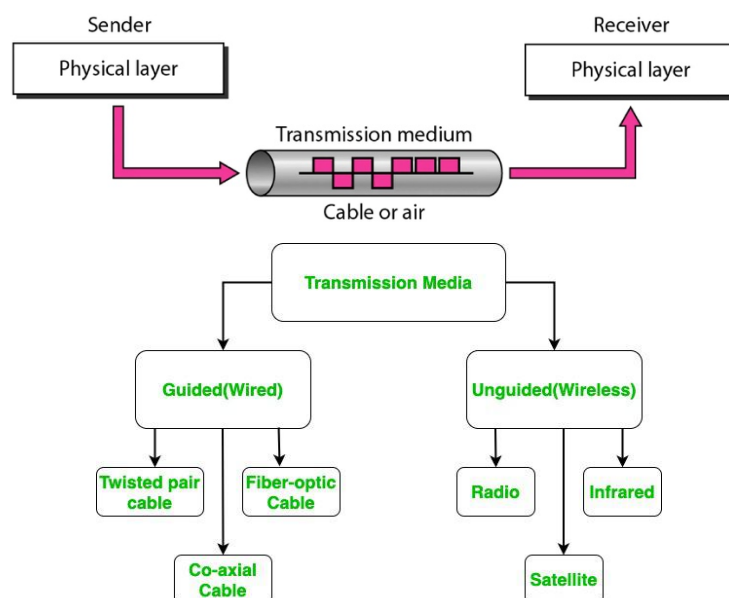
When a computer transmits data, the data is converted into electrical, optical, or electromagnetic signals. These signals travel through the transmission medium until they reach their destination.

Consider water flowing through a pipe. The pipe provides a pathway for water to move from one location to another. Similarly, transmission media provide pathways through which information travels between network devices.

Without transmission media, communication networks cannot exist.

Communication Through Transmission Media

Figure 7.1 *Transmission medium and physical layer*



The choice of transmission medium affects several network characteristics, including transmission speed, communication distance, installation cost, maintenance requirements, signal quality, and overall network performance.

Classification of Transmission Media

Transmission media can be broadly classified into two categories:

1. Guided Transmission Media
2. Unguided Transmission Media

The distinction between these categories lies in how signals travel from the sender to the receiver.

In guided media, signals travel through a physical pathway such as a cable. In unguided media, signals propagate through free space using electromagnetic waves.

Guided Transmission Media

Guided media, also known as wired media, use a physical conductor to guide signals from one device to another. The signal remains confined within the transmission path throughout its journey.

Because the path is controlled, guided media generally provide higher reliability, lower interference, and better security than wireless communication.

The most common forms of guided media are twisted pair cables, coaxial cables, and optical fiber cables.

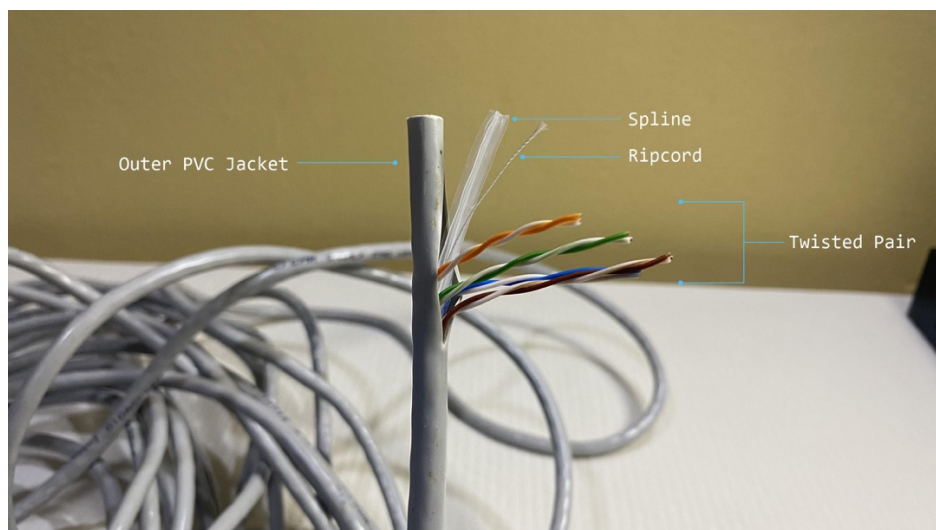
1. Twisted Pair Cable

The twisted pair cable is the most widely used transmission medium in computer networks. It consists of two insulated copper wires twisted together. The twisting helps reduce electromagnetic interference and signal crosstalk between adjacent wires.

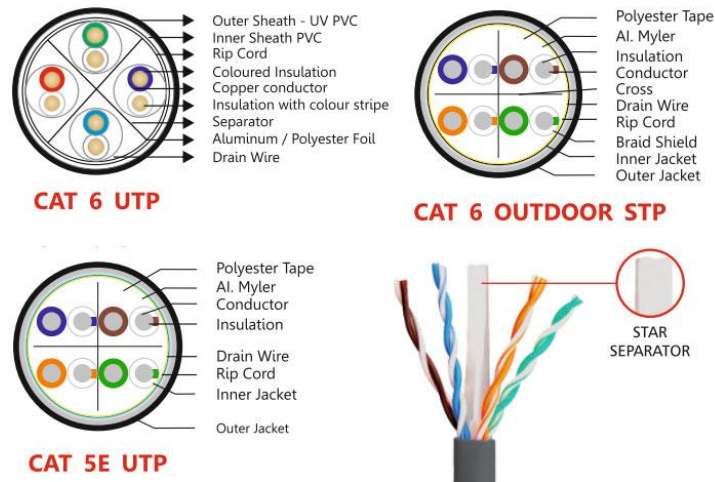
Twisted pair cables are commonly found in:

- Local Area Networks (LANs)
- Telephone systems
- Office networks
- Educational institutions

Structure of Twisted Pair Cable



A Cat 6 UTP cable is a standardized, unshielded twisted-pair Ethernet cable used for computer networks. It reliably supports high-speed data transfers up to 1 Gbps (Gigabit Ethernet) over distances up to 100 meters, operating at a frequency of 250 MHz.



A Cat 5 UTP cable is a standard networking cable used to connect computers, routers, and switches to a Local Area Network (LAN).

The twisting pattern is crucial because it minimizes the effects of external electrical noise.

Types of Twisted Pair Cables

Twisted pair cables are primarily classified into:

- **Unshielded Twisted Pair (UTP):** UTP cables do not contain additional shielding material. They are inexpensive, lightweight, and easy to install. Most Ethernet networks use UTP cables.
- **Shielded Twisted Pair (STP):** STP cables contain an additional metallic shield that provides protection against electromagnetic interference. They are commonly used in industrial environments where electrical noise is significant.

Advantages of Twisted Pair Cable

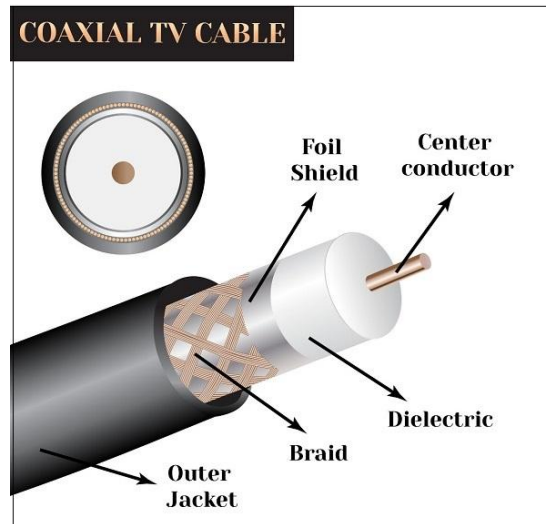
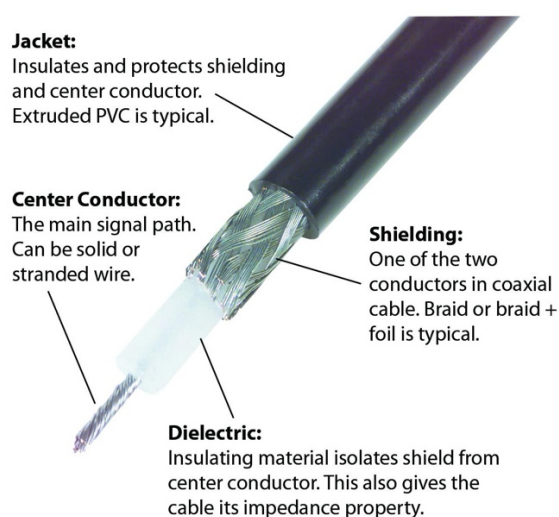
Twisted pair cables are economical and easy to install. They are flexible, lightweight, and widely supported by networking equipment. Their low cost makes them suitable for large-scale deployments.

Limitations of Twisted Pair Cable

As communication distance increases, signal attenuation becomes more significant. Twisted pair cables are also more susceptible to interference than optical fiber and generally support lower bandwidth.

2. Coaxial Cable

Coaxial cable was developed to provide better shielding and higher transmission performance than twisted pair cables. A coaxial cable consists of a central copper conductor surrounded by insulating material, a metallic shield, and an outer protective covering.



The metallic shield protects transmitted signals from external interference, resulting in improved communication quality.

Applications of Coaxial Cable

Coaxial cables are commonly used in cable television systems, broadband Internet connections, surveillance systems, and radio frequency communication. Historically, coaxial cables were widely used in Ethernet networks before twisted pair cables became dominant.

Advantages of Coaxial Cable

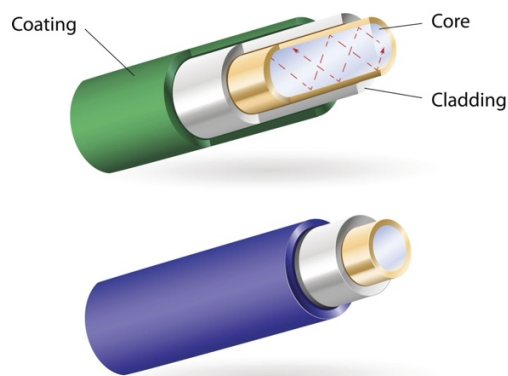
Coaxial cables offer higher bandwidth and better noise immunity than twisted pair cables. They support longer transmission distances and provide more stable signal quality.

Limitations of Coaxial Cable

Despite their advantages, coaxial cables are thicker, heavier, and more expensive. Installation and maintenance are also more complex.

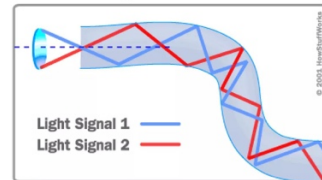
3. Optical Fiber Cable

As communication demands increased, traditional copper-based transmission media became insufficient. Modern applications such as cloud computing, high-definition streaming, artificial intelligence, and large-scale data centers require extremely high transmission speeds. Optical fiber was developed to meet these requirements. Unlike copper cables that use electrical signals, optical fiber transmits information using light pulses.



How Does an Optical Fiber Transmit Light?

The light in a fiber-optic cable travels through the core (hallway) by constantly bouncing from the cladding (mirror-lined walls), a principle called total internal reflection. Because the cladding does not absorb any light from the core, the light wave can travel great distances.



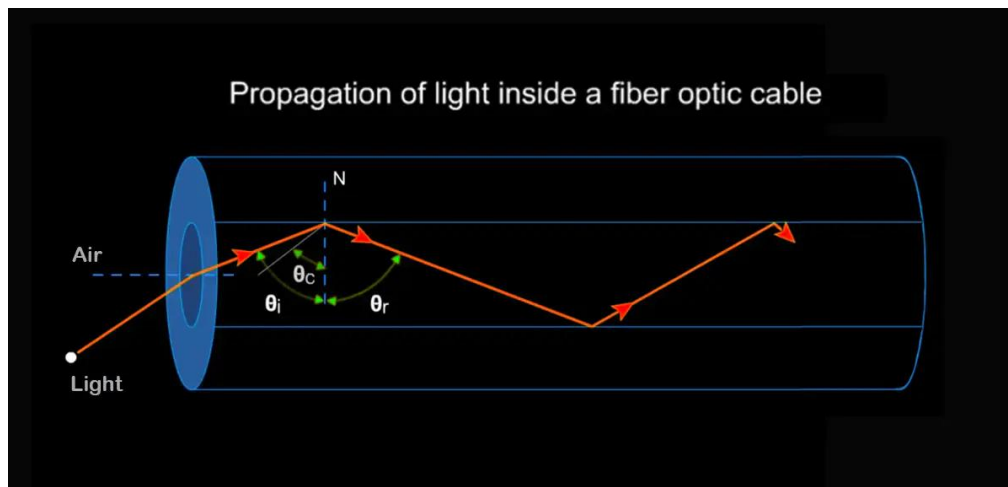
An optical fiber consists of three major components:

- The core carries light signals.
- The cladding surrounds the core and reflects light back into the core through total internal reflection.
- The protective jacket protects the fiber from physical damage.

Principle of Operation

Optical fiber communication relies on the principle of total internal reflection.

When light enters the fiber core at an appropriate angle, it continuously reflects within the core and travels long distances with minimal signal loss.



This mechanism allows optical fibers to transmit enormous amounts of data over very long distances.

Advantages of Optical Fiber

Optical fiber provides extremely high bandwidth, low attenuation, immunity to electromagnetic interference, enhanced security, and long-distance communication capabilities. Most modern Internet backbone networks rely on optical fiber technology.

Limitations of Optical Fiber

Although optical fiber offers superior performance, it is more expensive than copper-based alternatives. Installation requires specialized equipment and trained personnel.

Unguided Transmission Media

Unlike guided media, unguided media do not require a physical conductor. Instead, signals travel through free space using electromagnetic waves. Wireless communication has become increasingly important because it enables mobility and flexibility.

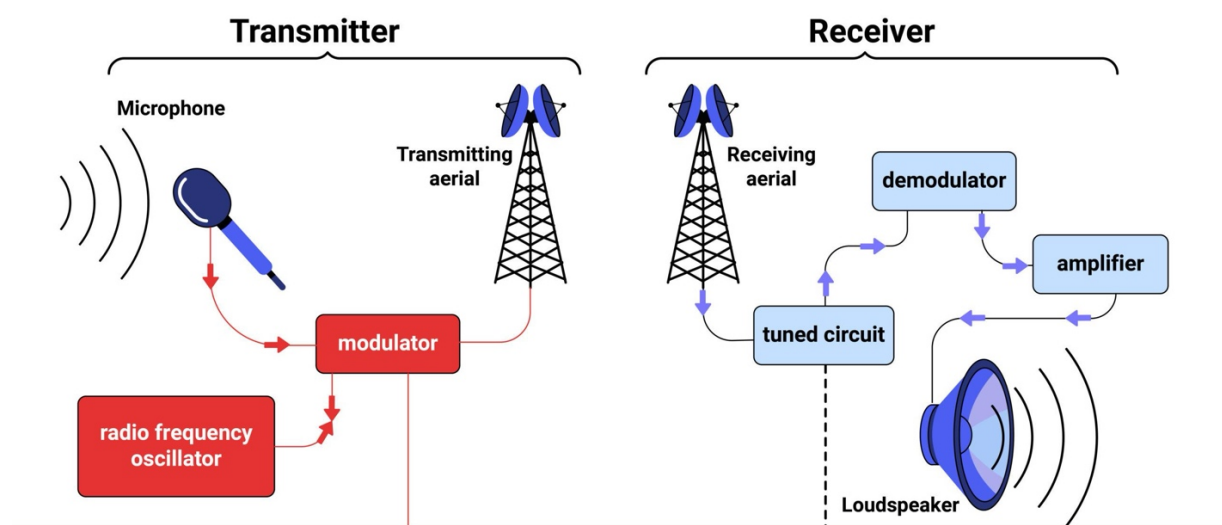
Mobile phones, Wi-Fi networks, satellite communication, and radio broadcasting all rely on unguided transmission media.

1. Radio Waves

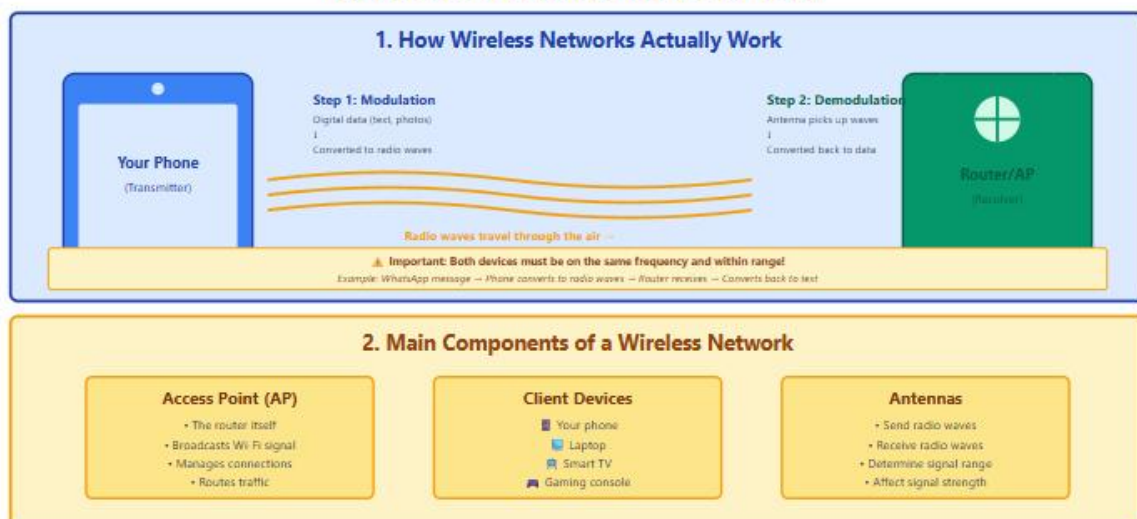
Radio waves can travel long distances and penetrate buildings effectively.

They are widely used in:

- FM radio broadcasting
- Television broadcasting
- Mobile communication
- Wireless networking



How Wireless Networks Actually Work



Because radio waves can travel in multiple directions, they are suitable for broad-area communication.

2. Microwaves

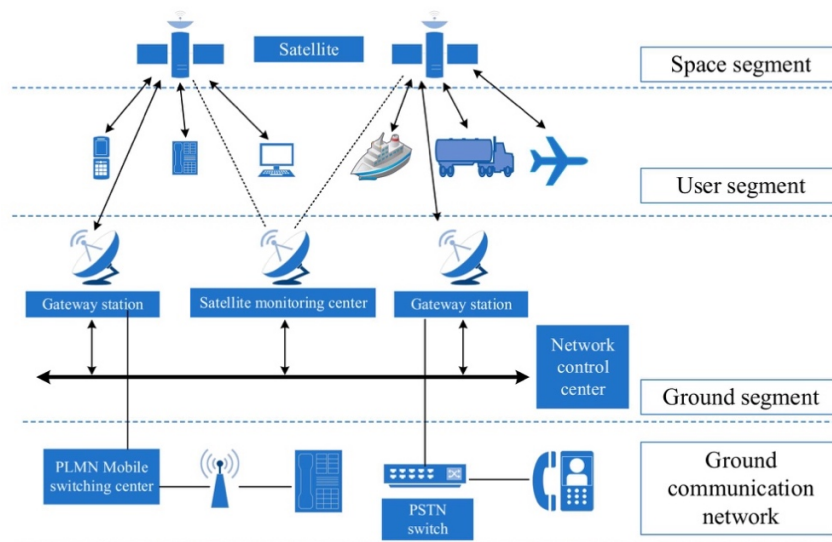
Microwave communication uses higher frequencies than traditional radio waves. Microwave transmission generally requires line-of-sight communication between transmitting and receiving antennas.

Applications include cellular communication, satellite communication, and long-distance telecommunications links.



3. Satellite Communication

Satellite communication extends wireless communication beyond terrestrial limitations. Communication signals are transmitted from Earth to a satellite and then relayed back to another location.



Satellite communication enables global connectivity and supports television broadcasting, navigation systems, military communication, and disaster recovery operations.

Understanding Network Performance

Having a communication medium alone does not guarantee effective communication. Engineers must also evaluate how efficiently the network performs. Network performance refers to the ability of a network to deliver data accurately, quickly, reliably, and efficiently.

Performance determines the quality of user experience and directly influences the effectiveness of communication systems.

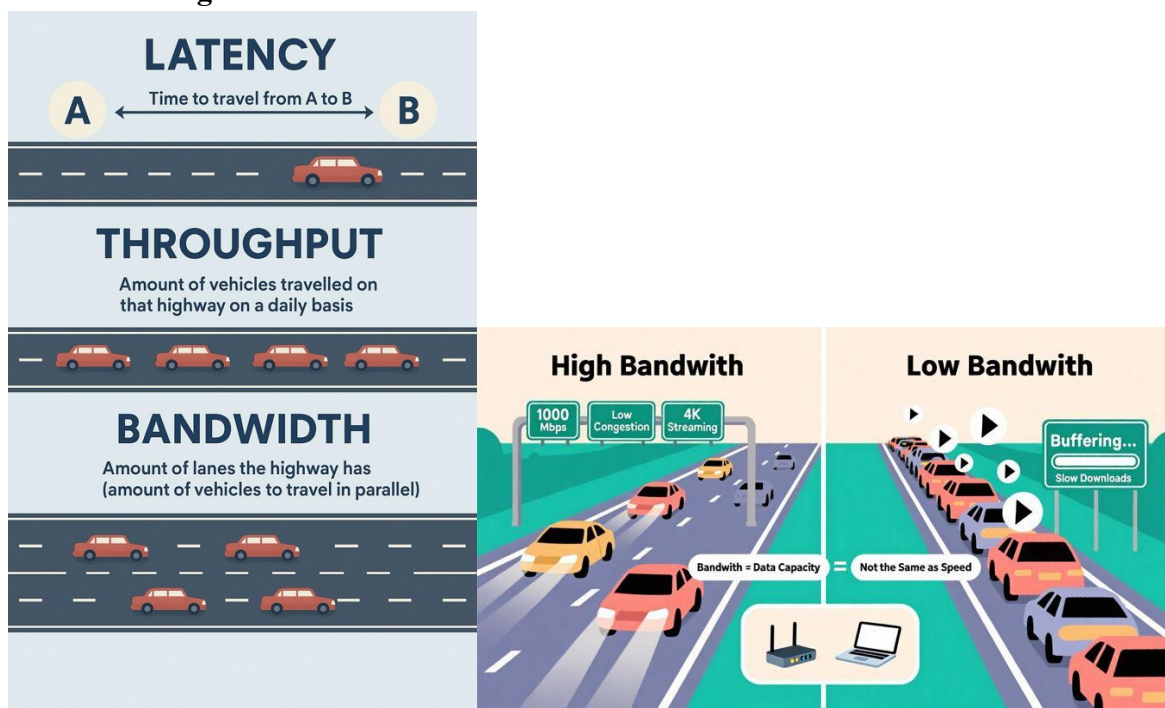
Bandwidth: Bandwidth is one of the most important performance metrics in networking. Bandwidth represents the maximum amount of data that can be transmitted through a communication channel within a given period.

It is usually measured in:

- bits per second (bps)
- kilobits per second (Kbps)
- megabits per second (Mbps)
- gigabits per second (Gbps)

A network with higher bandwidth can carry more information simultaneously.

Understanding Bandwidth



Bandwidth can be compared to the width of a highway. A wider highway accommodates more vehicles at the same time. Similarly, higher bandwidth accommodates more data.

Throughput: While bandwidth represents theoretical capacity, throughput represents actual performance. Throughput refers to the amount of data successfully transmitted over the network in a given period. Due to congestion, interference, protocol overhead, and hardware limitations, throughput is often lower than bandwidth.

For example, a network advertised as 100 Mbps may deliver an actual throughput of 80 Mbps under normal operating conditions. Throughput is calculated as below if number of frames and frame size are given

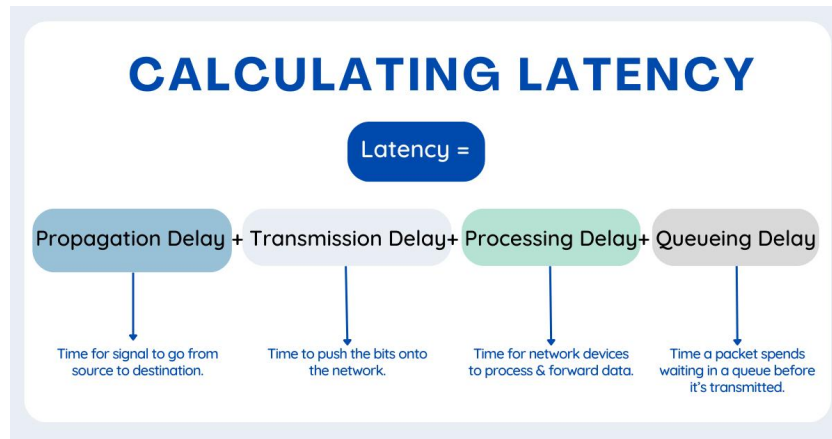
$$\text{Throughput} = (\text{Number of Frames} \times \text{Frame Size (bits)}) / \text{Time}$$

Or if just the amount of data transmitted in a period of time is given,

$$\text{Throughput} = \text{Total Data Successfully Transmitted} / \text{Time Taken}$$

Latency: Latency refers to the time required for data to travel from the sender to the receiver. It is usually measured in milliseconds (ms).

Low latency is essential for applications such as online gaming, video conferencing, remote surgery, and autonomous vehicles.



A network may have high bandwidth but still perform poorly if latency is excessive.

Jitter: Jitter refers to variations in packet arrival times. In applications such as voice calls and video conferencing, packets should arrive at regular intervals. Significant variation can cause audio distortion and video interruptions.

Modern communication systems implement buffering mechanisms to reduce the impact of jitter.

Reliability: Reliability measures the ability of a network to operate consistently without failures.

A reliable network ensures:

- Accurate data delivery
- Minimal downtime
- Stable performance
- Effective fault recovery

Reliability is particularly important in banking systems, healthcare networks, industrial automation systems, and telecommunications infrastructure.

Transmission media provide the physical foundation upon which all network communication occurs. Guided media such as twisted pair cables, coaxial cables, and optical fibers provide controlled communication paths, while unguided media such as radio waves, microwaves, and satellites enable wireless communication. The selection of a transmission medium significantly influences network performance.

Network performance is evaluated through metrics such as bandwidth, throughput, latency, jitter, and reliability. These parameters determine how efficiently data moves through the network and directly affect user experience. Together, transmission media and performance characteristics form the practical foundation of the Physical Layer and play a crucial role in the design and operation of modern computer networks.

Switching

Switching is a technique used in computer networks (mostly in LANs) to transfer data from a source device to a destination device through intermediate network devices (switches, routers, exchanges, etc.). Since it is impractical to provide a dedicated communication link between every pair of devices, switching helps share network resources efficiently.

Switching is the process of receiving data from one device and forwarding it through the appropriate path to the intended destination.

Need for Switching

- Direct links between all devices are expensive and impractical.
- Multiple users need to share communication resources.
- Switching improves network scalability and efficiency.
- It enables communication over long distances through intermediate nodes.

Types of Switching

1. Circuit Switching

A dedicated communication path is established between the sender and receiver before data transmission begins.

Phases:

1. Connection Establishment
2. Data Transfer
3. Connection Release

Characteristics:

- Dedicated path throughout communication.
- Fixed bandwidth.
- No delay due to route selection after setup.
- Suitable for voice calls.

Example: Traditional telephone network.

Advantages:

- Guaranteed bandwidth.
- Consistent delay.

Disadvantages:

- Resource wastage when the circuit is idle.
- Setup time required.

2. Message Switching

The entire message is treated as a single unit and transmitted from one node to another using the **Store-and-Forward** technique.

Characteristics:

- No dedicated path.
- Each intermediate node stores the complete message before forwarding.
- Large storage requirement.

Advantages:

- Efficient utilization of network resources.

- No need for dedicated circuits.

Disadvantages:

- High delay due to storage and forwarding.
- Not suitable for real-time communication.

Example: Telegraph systems.

3. Packet Switching

The message is divided into smaller units called **packets**. Each packet is transmitted independently through the network.

Characteristics:

- No dedicated path required.
- Better utilization of bandwidth.
- Basis of the Internet.

Advantages:

- Efficient bandwidth usage.
- Supports many users simultaneously.
- Reliable and scalable.

Disadvantages:

- Variable delay.
- Packets may arrive out of order.

Example: Internet communication.

Types of Packet Switching

a) Datagram Packet Switching

- Each packet is routed independently.
- Different packets may take different paths.
- No setup phase.

Example: Internet Protocol (IP).

b) Virtual Circuit Packet Switching

- A logical path is established before transmission.
- All packets follow the same route.
- Combines advantages of circuit and packet switching.

Example: Frame Relay, ATM.

Comparison of Switching Techniques

Feature	Circuit Switching	Message Switching	Packet Switching
Dedicated Path	Yes	No	No
Setup Required	Yes	No	No (Datagram)
Data Unit	Continuous Stream	Entire Message	Packets
Delay	Low after setup	High	Moderate
Resource Utilization	Poor	Better	Best
Real-Time Communication	Suitable	Not Suitable	Suitable

Feature	Circuit Switching	Message Switching	Packet Switching
Example	Telephone Network	Telegraph	Internet

Simple Illustration

Circuit Switching

A ===== B
(Dedicated Path)

Message Switching

A -> Node1 -> Node2 -> B

(Whole message stored and forwarded)

Packet Switching

Packet1 -> Path1

Packet2 -> Path2

Packet3 -> Path3

(All reassembled at destination)

SWITCHING IN COMPUTER NETWORKS

1. INTRODUCTION

Switching is the process of receiving data from one device and forwarding it through the appropriate path to the intended destination using intermediate network devices (switches, routers, exchanges, etc.).

NEED FOR SWITCHING

- Direct links between all devices are expensive and impractical.
- Multiple users need to share communication resources.
- Switching improves network scalability and efficiency.
- It enables communication over long distances through intermediate nodes.

Instead of direct links between all pairs...

... switching reduces links and cost.

2. CIRCUIT SWITCHING

A dedicated communication path is established between the sender and receiver before data transmission begins.

PHASES

- 1 Connection Establishment
- 2 Data Transfer
- 3 Connection Release

Characteristics

- Dedicated path throughout communication.
- Fixed bandwidth.
- No delay due to route selection after setup.
- Suitable for voice calls.

Example: Traditional telephone network

Advantages

- Guaranteed bandwidth
- Consistent delay

Disadvantages

- Resource wastage when the circuit is idle
- Setup time required

3. MESSAGE SWITCHING

The entire message is treated as a single unit and transmitted from one node to another using the Store-and-Forward technique.

Characteristics

- No dedicated path.
- Each intermediate node stores the complete message before forwarding.
- Large storage requirement.

Example: Telegraph systems

Advantages

- Efficient utilization of network resources
- No need for dedicated circuits

Disadvantages

- High delay due to storage and forwarding
- Not suitable for real-time communication

4. PACKET SWITCHING

The message is divided into smaller units called packets. Each packet is transmitted independently through the network.

Characteristics

- No dedicated path required.
- Better utilization of bandwidth.
- Basis of the Internet.

Example: Internet communication

Advantages

- Efficient bandwidth usage
- Supports many users simultaneously
- Reliable and scalable

Disadvantages

- Variable delay
- Packets may arrive out of order

TYPES OF PACKET SWITCHING

a) Datagram Packet Switching

- Each packet is routed independently.
- Different packets may take different paths.
- No setup phase.

Example: Internet Protocol (IP)

b) Virtual Circuit Packet Switching

- A logical path is established before transmission.
- All packets follow the same route.
- Combines advantages of circuit and packet switching.

Example: Frame Relay, ATM

COMPARISON OF SWITCHING TECHNIQUES

Feature	Circuit Switching	Message Switching	Packet Switching
Dedicated Path	Yes	No	No
Setup Required	Yes	No	No (Datagram)
Data Unit	Continuous Stream	Entire Message	Packets
Delay	Low after setup	High	Moderate
Resource Utilization	Poor	Better	Best
Real-Time Communication	Suitable	Not Suitable	Suitable
Example	Telephone Network	Telegraph	Internet

SIMPLE ILLUSTRATION

Circuit Switching

(Dedicated Path)

Message Switching

(Whole message stored and forwarded)

Packet Switching

(All packets take different paths and reassembled at destination)

EXAM DEFINITION (2 MARKS)

Switching is the process of forwarding data from a source to a destination through intermediate network devices. The main switching techniques are Circuit Switching, Message Switching, and Packet Switching.

16-MARK ANSWER POINTS

1. Introduction to Switching
2. Need for Switching
3. Circuit Switching – Working, Advantages, Disadvantages
4. Message Switching – Working, Advantages, Disadvantages
5. Packet Switching – Working, Advantages, Disadvantages
6. Datagram and Virtual Circuit Packet Switching
7. Comparison Table
8. Applications and Conclusion

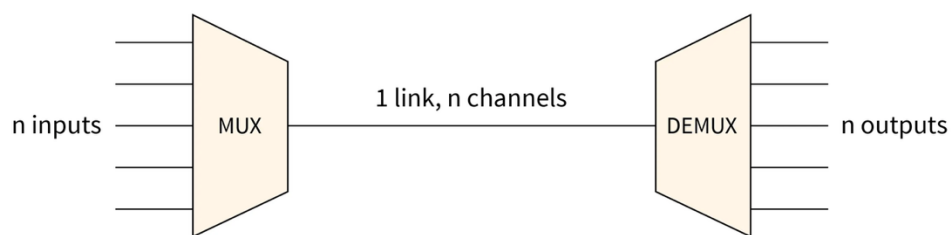
Multiplexing in Computer Networks

Multiplexing is one of the fundamental concepts in data communications and computer networks. As communication networks grew larger, it became impractical and expensive to provide a separate transmission medium for every user or device. Multiplexing was developed as a solution to allow multiple users to share a single high-capacity communication channel simultaneously.

The term **multiplexing** is derived from the words *multiple* and *plexing*, meaning the process of combining multiple signals into a single transmission path.

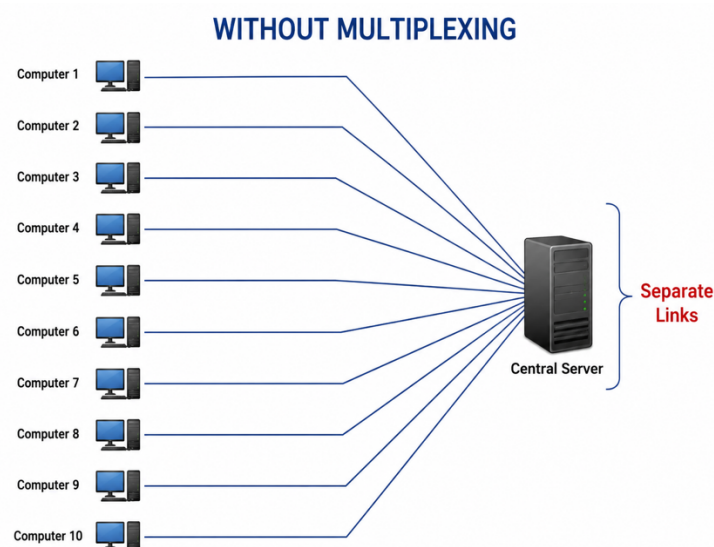
In a multiplexing system, several independent signals generated by different users are combined at the sender side by a device called a **Multiplexer (MUX)**. The combined signal is transmitted through a shared communication medium. At the receiving end, another device called a **Demultiplexer (DEMUX)** separates the composite signal into its original individual signals and delivers them to the appropriate destinations.

Basic Multiplexing Model



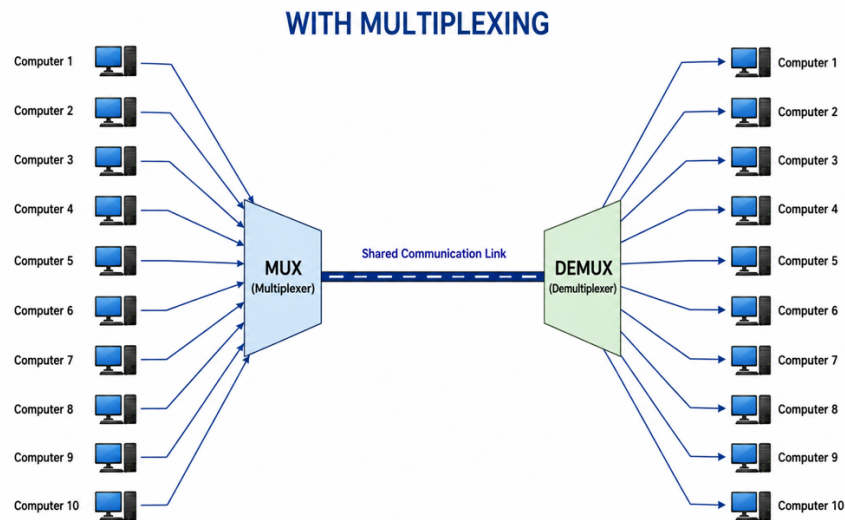
Need for Multiplexing

Consider a situation where ten computers need to transmit data to a central server.



Each computer requires an independent communication channel. This increases:

- Infrastructure cost
- Cable requirements
- Maintenance complexity
- Network management overhead



A single communication link is shared among all users, resulting in better bandwidth utilization and reduced cost.

PRINCIPLE OF MULTIPLEXING

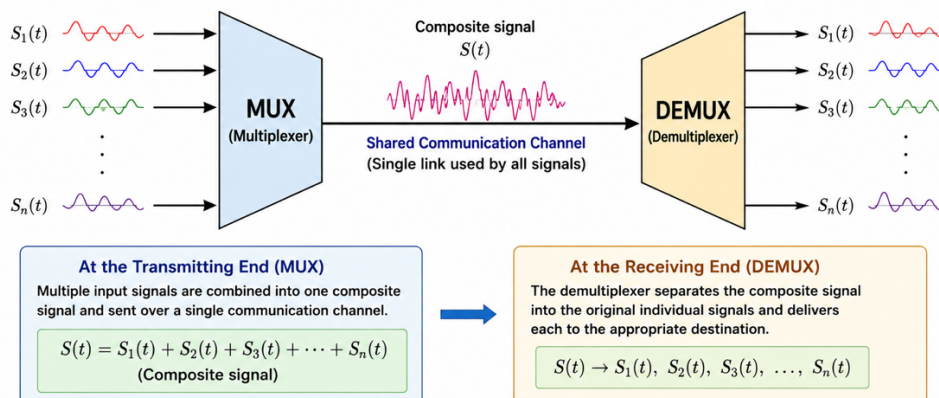
Suppose there are n input signals:

$$S_1(t), S_2(t), S_3(t), \dots, S_n(t)$$

The multiplexer combines these signals into a single composite signal:

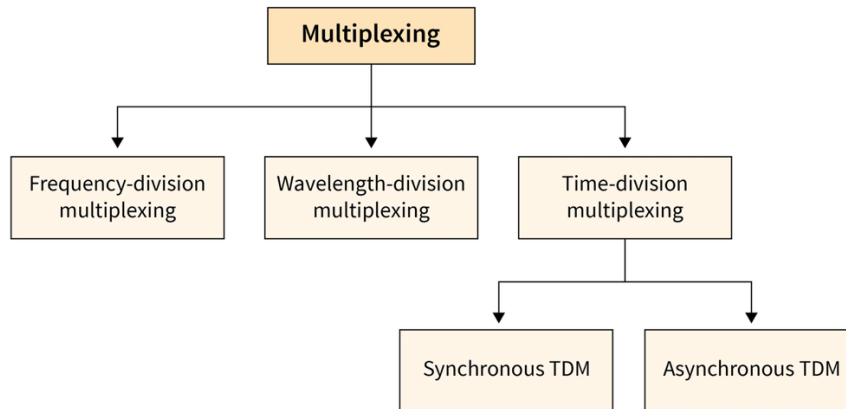
$$S(t) = S_1(t) + S_2(t) + S_3(t) + \dots + S_n(t) = \sum_{i=1}^n S_i(t)$$

The exact method of combining depends on the multiplexing technique used.



Classification of Multiplexing

Multiplexing techniques are broadly classified into:



Frequency Division Multiplexing (FDM)

Frequency Division Multiplexing is an analog multiplexing technique in which the available channel bandwidth is divided into several non-overlapping frequency bands.

Each user is assigned a unique carrier frequency.

Working Principle

Suppose three users wish to transmit simultaneously.

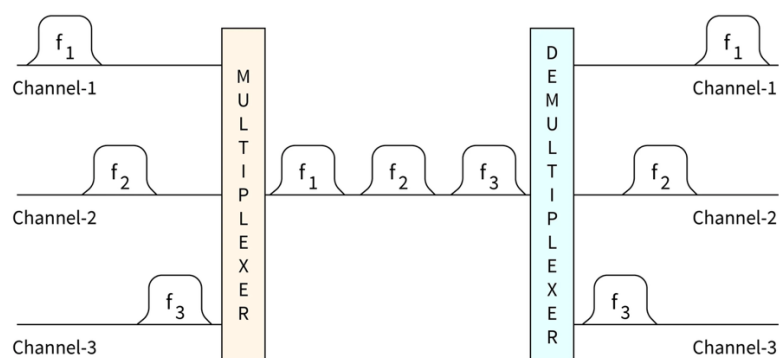
Channel 1 --> f_1

Channel 2 --> f_2

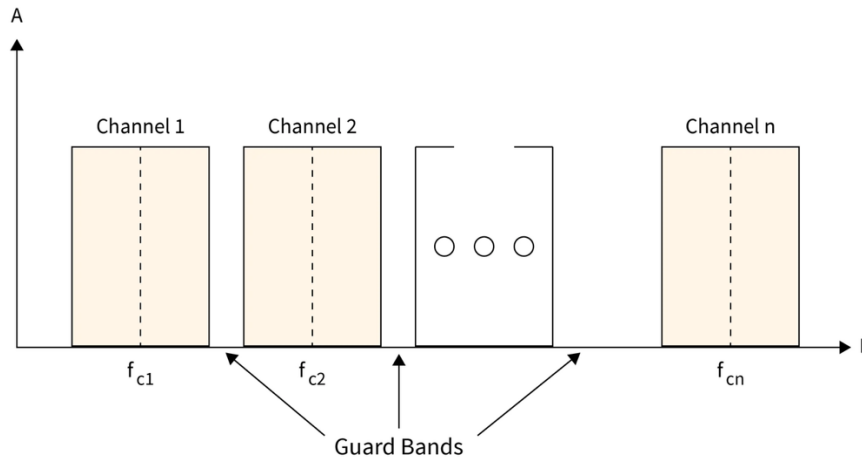
Channel 3 --> f_3

Each signal modulates a different carrier frequency.

The resulting spectrum appears as:

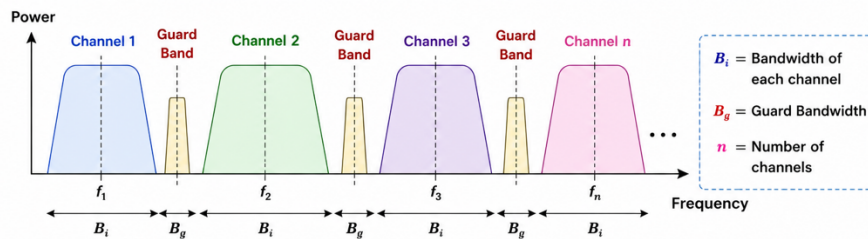


Guard bands are inserted between adjacent channels to prevent interference.



FDM BANDWIDTH FORMULA

In Frequency Division Multiplexing (FDM), the available bandwidth is divided into several **non-overlapping** frequency bands. Each channel is separated by a guard band to avoid interference.



Total Bandwidth Required

$$B_T = nB_i + (n - 1)B_g$$

The total bandwidth is equal to the sum of the bandwidths of all channels plus the guard bands between them.

Where,

B_T = Total bandwidth required
 B_i = Bandwidth of each channel
 B_g = Guard bandwidth
 n = Number of channels

Numerical Example

EXAMPLE: FIND TOTAL BANDWIDTH

A communication system supports 10 channels.

GIVEN

Number of channels
 $n = 10$

Bandwidth of each channel
 $B_i = 4 \text{ kHz}$

Guard band
 $B_g = 1 \text{ kHz}$

Find:
 Total bandwidth (B_T)

SOLUTION

The total bandwidth required is given by the formula:

$$B_T = nB_i + (n - 1)B_g$$

Substitute the given values:

$$B_T = (10 \times 4) + (10 - 1) \times 1$$

$$B_T = (10 \times 4) + (9 \times 1)$$

$$B_T = 40 + 9$$

$$B_T = 49 \text{ kHz}$$



Therefore, a minimum bandwidth of **49 kHz** is required.

Applications of FDM

Radio broadcasting is the most common example.

88 MHz 92 MHz 96 MHz 100 MHz

|-----|-----|-----|

Station A Station B Station C

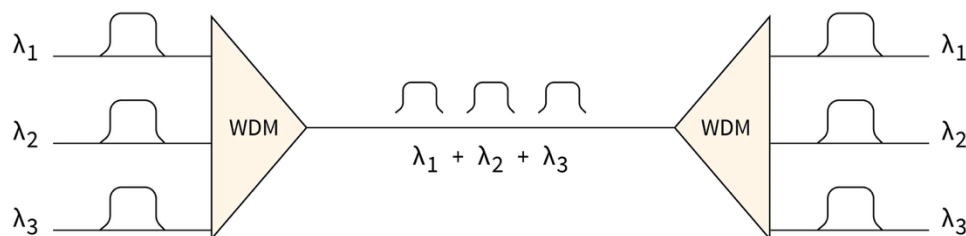
Each station occupies a unique frequency band.

Television broadcasting and cable TV networks also use FDM.

Wavelength Division Multiplexing (WDM)

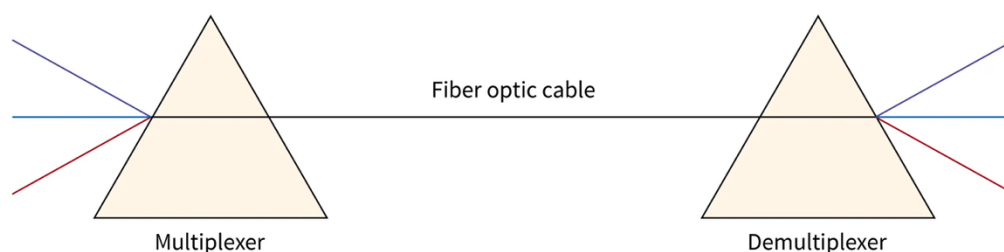
Wavelength Division Multiplexing is the optical equivalent of FDM. Wavelength Division Multiplexing (WDM) is used in Fiber optics to increase the capacity of a single Fiber.

Instead of frequencies, different wavelengths of light are used to carry different signals through the same optical Fiber.



Each wavelength carries independent information.

- In Wavelength Division Multiplexing, various light waves from different sources are merged into a composite light signal, and then transmitted across the channel to the receiver.
- It is an analog multiplexing approach and very much similar to FDM.
- Using multiplexers, optical signals from many sources are combined to generate a larger range of light. The demultiplexer differentiates the signals at the receiving end before sending them to their respective destinations.
- In WDM, Prisms are used to combine and split light waves. This prism assists in bending the light beam according to the angle of incidence and frequency of the light.
- Prism primarily serves as the multiplexer in the WDM technology. It then combines the multiple optical signals to create a composite signal, which is subsequently transmitted through an optical fiber connection.



Advantages

WDM significantly increases Fiber capacity because multiple optical channels share the same Fiber simultaneously.

Applications

- Fiber optic communication
- Internet backbone networks
- Submarine optical cables
- Long-distance telecommunication

Time Division Multiplexing (TDM)

Time Division Multiplexing is a digital multiplexing technique in which multiple users share the same channel by transmitting in different time slots.

Instead of dividing bandwidth, TDM divides time.

Working Principle

Suppose there are four users.

Each user is assigned a time slot.

Frame 1

| U1 | U2 | U3 | U4 |

Frame 2

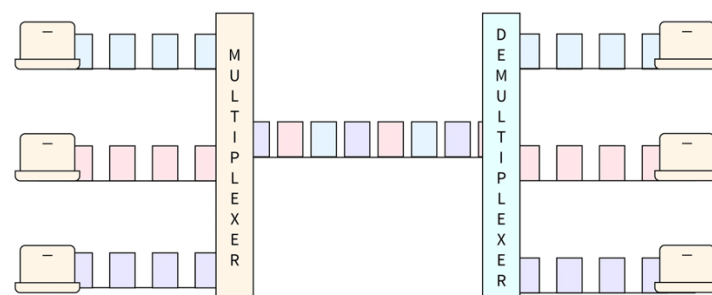
| U1 | U2 | U3 | U4 |

Frame 3

| U1 | U2 | U3 | U4 |

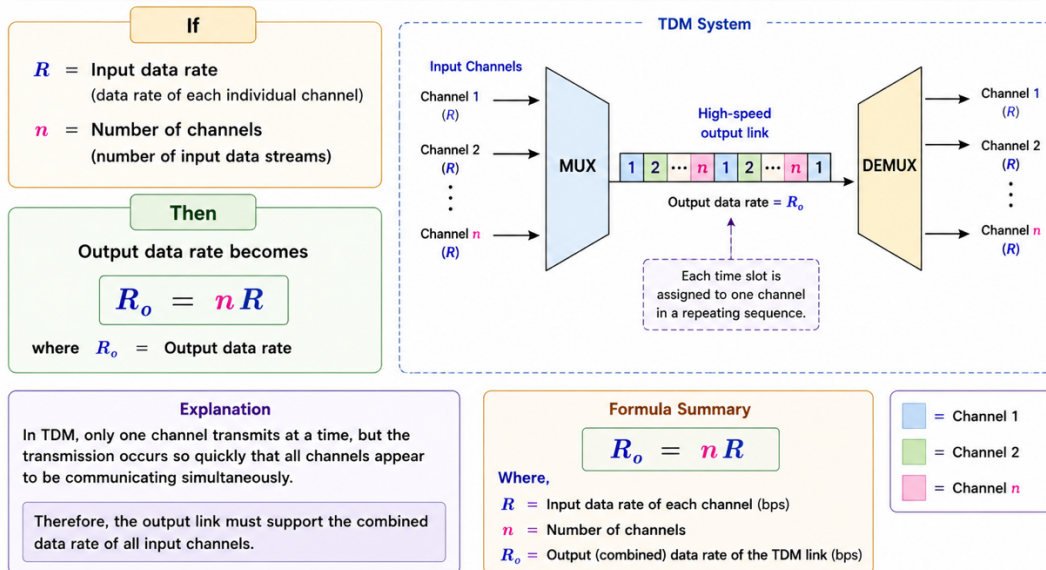
Only one user transmits at a time.

However, transmission occurs so rapidly that it appears simultaneous.



TDM DATA RATE FORMULA

In **Time Division Multiplexing (TDM)**, multiple input data streams share the same communication channel by transmitting in different time slots.



Numerical Example

Four devices each transmit at 100kbps. Find output rate.

Solution

$$R_o = nR$$

$$R_o = 4 \times 100$$

$$R_o = 400 \text{ kbps}$$

Hence, the multiplexed line must support **400 kbps**.

Frame Duration

If $R_o = 400 \text{ kbps}$ and one frame contains 4 bits, $T_f = \{4\} / \{400000\}$

$$T_f = 10 \mu\text{s}$$

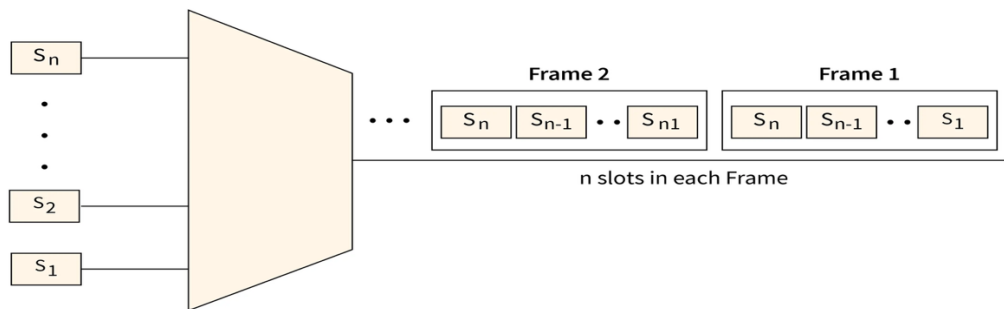
Thus one complete frame is transmitted every 10 microseconds.

There are generally two types of Time Division Multiplexing

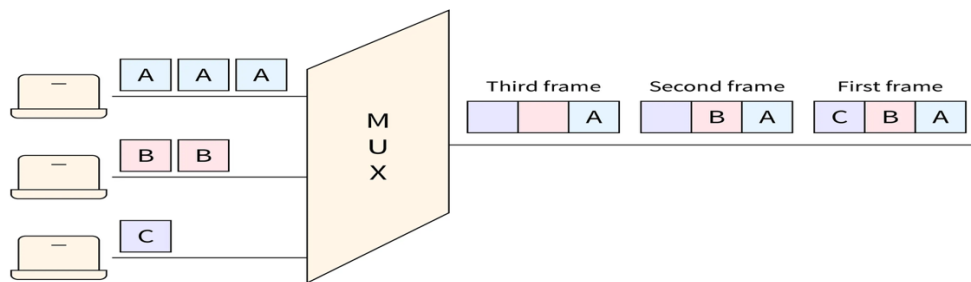
1. Synchronous TDM

- Synchronous Time Division Multiplexing is a sort of Time Division Multiplexing in which the input frame already has a slot in the output frame. Frames are groups of time slots. A single frame is comprised of one cycle of time slots.
- Whenever its time slot arrives, each device sends data to the link. Thus, each device is given control one at a time.

- Suppose there are 'n' sending devices; then there will be 'n' time slots simultaneously, which means a one-time slot for each device.



- Synchronous TDM is inefficient because if there is no data to convey in the input frame, a slot in the output frame remains vacant.
- In synchronous Time Division Multiplexing, the synchronous bit must be mentioned at the start of each frame.
- ISDN multiplexing, T-1 multiplexing, and SONET multiplexing are the most common Synchronous TDM.



Advantages of Synchronous Time Division Multiplexing

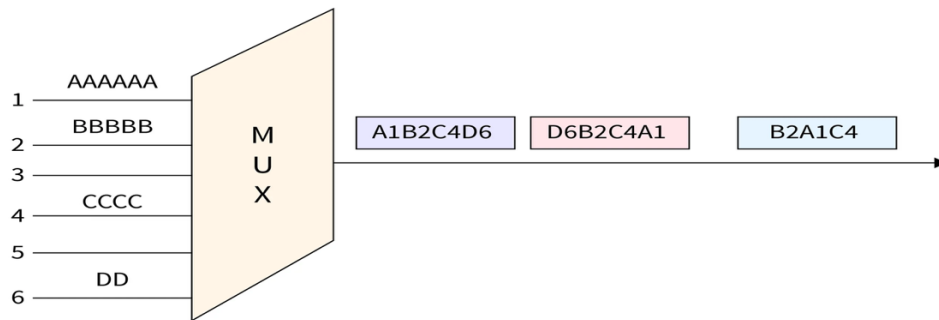
- Synchronous Time Division Multiplexing is simple to implement.
- This strategy ensures good and reliable performance.

Disadvantages of Synchronous Time Division Multiplexing

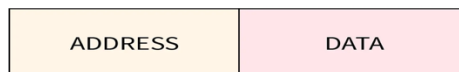
- The channel's capacity is not fully utilized because empty slots with no data are also conveyed.
- The transmission medium's speed should be greater than the overall speed of the input lines.

2. Asynchronous Time Division Multiplexing

- Asynchronous Time Division Multiplexing, sometimes also known as Statical Time Division Multiplexing, is a type of multiplexing in which time slots are not set but are allocated dynamically to optimize bandwidth efficiency.
- There is no concept of predefined slots; instead, slots are assigned dynamically based on demand.



- Asynchronous Time Division multiplexer receives incoming data streams and generates a frame with no empty slots.
- There are n input lines and m slots in this Multiplexing; therefore, always $(m < n)$.
- The overall speed of the input lines in asynchronous TDM can be more than the channel capacity.
- The distinction between Asynchronous TDM and Synchronous TDM is that many slots in Synchronous TDM are unused, whereas slots in Asynchronous TDM are entirely utilized. This results in a shorter transmission time and more efficient channel utilization.
- Each slot in Asynchronous TDM comprises an address element that identifies the data source.



Advantages of Asynchronous Time Division Multiplexing

- The transmission capacity is used efficiently in this multiplexing.

Disadvantages of Asynchronous Time Division Multiplexing

- Since there are no dedicated slots for each user, the buffer address information is also required.
- The waiting time is not fixed.

Code Division Multiplexing (CDM/CDMA)

Code Division Multiplexing allows all users to transmit simultaneously using the same frequency band.

Instead of assigning different frequencies or time slots, each user receives a unique code sequence.

Working Principle

User 1 --> Code A

User 2 --> Code B

User 3 --> Code C

All signals are transmitted together.

Composite Signal = Signal A + Signal B + Signal C

The receiver extracts the desired signal using the corresponding code.

Mathematical Representation

Let $m_i(t)$ be the message signal and $c_i(t)$ be its unique code.

Then $s(t) = \sum_{i=1}^n m_i(t)c_i(t)$

At the receiver, correlation techniques recover the intended signal.

Applications

CDMA is extensively used in:

- 3G mobile networks
- GPS systems
- Satellite communication
- Military communication systems

Comparison of Multiplexing Techniques

Feature	FDM	WDM	TDM	STDM	CDM
Basis	Frequency	Wavelength	Time	Dynamic Time	Code
Signal Type	Analog	Optical	Digital	Digital	Digital
Simultaneous Transmission	Yes	Yes	No	No	Yes
Synchronization	Not Required	Not Required	Required	Required	Required
Efficiency	Good	Very High	Moderate	High	Very High
Typical Application	Radio	Fiber Optics	PCM Systems	Computer Networks	Mobile Networks

Advantages of Multiplexing

Multiplexing improves bandwidth utilization by allowing several users to share the same communication channel. It reduces infrastructure cost, decreases the number of physical communication links, increases transmission efficiency, and enables large-scale communication networks to operate economically. Modern Internet, cellular networks, fiber-optic systems, and satellite communications would not be feasible without multiplexing.

Limitations of Multiplexing

Multiplexing introduces additional complexity because multiplexers and demultiplexers are required at both ends. Certain techniques require precise synchronization. A failure in the shared communication link can affect multiple users simultaneously. Interference and crosstalk may also occur if channels are not properly separated.